



LINEAR REGRESSION

PREDICTING BLUEBIKES RENTAL AND SOLAR RADIATION

Train/Test Split

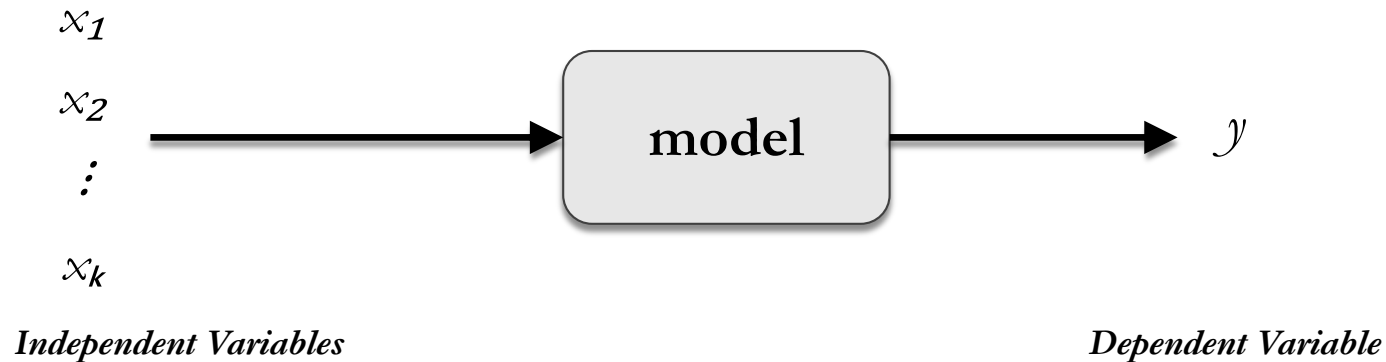


Reminders



- Fuqua Check-in
- Form your team of 3

Predictive Model



- We want to predict the value of one* variable called the **dependent variable** (synonyms: the target variable, labels)...
- using values of the other variables, called the **independent variables** (synonyms: explanatory variable, predictor, feature)

*: You can have multiple dependent variables. We will focus on a single dependent variable in this course.

Predicting **BLUEbikes** Rentals

- According to Wikipedia:

“Bluebikes... is a bicycle sharing system in the Boston metropolitan area. As of 2024, the system had deployed 520 stations with a fleet of over 5,500 bikes in the 10 municipalities it served.”

- Bluebikes' System Data:

- Bluebikes makes available anonymized data for every trip taken.

<https://bluebikes.com/system-data>; <https://s3.amazonaws.com/hubway-data/202401-bluebikes-tripdata.zip>

ride_id	rideable_type	started_at	ended_at	start_station_name	start_station_id	end_station_name	end_station_id	start_lat	start_lng	end_lat	end_lng	member_casual
D2F4A4783B230A84	electric_bike	2024-01-31 12:16:49	2024-01-31 12:21:02	Ames St at Main St	M32037	Central Square at Mass Ave / Essex St	M32011	42.362357	-71.088163	42.365070	-71.103100	member
D305CEFFD4558633	classic_bike	2024-01-12 08:14:16	2024-01-12 08:19:48	Ames St at Main St	M32037	Central Square at Mass Ave / Essex St	M32011	42.362500	-71.088220	42.365070	-71.103100	member
02009BB4EBA0D1F6	electric_bike	2024-01-29 15:00:05	2024-01-29 15:05:47	One Memorial Drive	M32053	Kennedy-Longfellow School 158 Spring St	M32065	42.361697	-71.080273	42.369553	-71.085790	member
04C230C1C39071F7	classic_bike	2024-01-09 16:33:40	2024-01-09 17:00:41	Ames St at Main St	M32037	Brookline Town Hall	K32005	42.362500	-71.088220	42.333765	-71.120464	member
CEAFE67E28B43852	classic_bike	2024-01-23 10:19:21	2024-01-23 10:31:39	Mass Ave T Station	C32063	Chinatown T Stop	D32019	42.341356	-71.083370	42.352409	-71.062679	member
07C37921B6834977	classic_bike	2024-01-26 12:35:20	2024-01-26 12:54:06	St. Alphonsus St at Tremont St	B32063	Stuart St at Charles St	D32011	42.333293	-71.101246	42.351146	-71.066289	member

What to predict?

- What do you want to do with this data?
- What business decisions can this data help with?
- How to do it?

Predict Total* Hourly BlueBikes Rentals

For example: “What is the number of BlueBike rentals tomorrow between 5-6 p.m.?”

- What can we do with a model that predicts total hourly rentals?
 - Always ask: Why would we want to predict that? What’s the model’s decision-relevance?
- Say we need to staff the customer service center differently when demand is high (say 500-1000 rentals per hour). What would you need to know to make the right staffing decision?
- You may also want the **distribution** of rentals for tomorrow between 5-6pm, to calculate the probabilities of rentals in certain ranges.

* *We could predict rentals at each station instead, which would be useful for knowing how to rebalance bikes.*

First Predictor

- Aggregate to obtain number of rentals per hour. How do you predict the rental need for the next hour?

185. 208. 92. 32. 6. 14. 25. 30. 49. 105. 94. 139. 183. 187.
251. 285. 214. 175. 126. 86. 76. 67. 28. 20. 17. 12. 7. 4.
5. 38. 104. 252. 462. 262. 163. 184. 214. 226. 249. 339. 493. 583.
420. 236. 152. 95. 40. 40. 14. 9. 7. 3. 8. 53. 128. 352.
645. 368. 222. 214. 254. 235. 272. 373. 616. 764. 477. 279. 185. 108.
70. 34. 18. 10. 7. 6. 10. 49. 138. 358. 678. 387. 225. 260.
304. 277. 324. 374. 557. 708. 470. 292. 172. 110. 61. 37. 15. 8.

- Take Average! $\bar{y} = 540.96 \rightarrow$ *baseline model*
- Is this a good prediction? Why?
- How to measure/quantify the *baseline model*'s performance?
- Sum of Squared Total (SST):

$$SST = \sum_i (y_i - \bar{y})^2 = (y_1 - \bar{y})^2 + (y_2 - \bar{y})^2 + \cdots (y_n - \bar{y})^2 = 2,214 \text{ million}$$

Better Prediction than Simple Average?

| ride_id | rideable_type | started_at | ended_at | start_station_name | start_station_id | end_station_name | end_station_id | start_lat | start_lng | end_lat | end_lng | member_casual |
|------------------|---------------|---------------------|---------------------|--------------------------------|------------------|---|----------------|-----------|------------|-----------|------------|---------------|
| D2F4A4783B230A84 | electric_bike | 2024-01-31 12:16:49 | 2024-01-31 12:21:02 | Ames St at Main St | M32037 | Central Square at Mass Ave / Essex St | M32011 | 42.362357 | -71.088163 | 42.365070 | -71.103100 | member |
| D305CEFFD4558633 | classic_bike | 2024-01-12 08:14:16 | 2024-01-12 08:19:48 | Ames St at Main St | M32037 | Central Square at Mass Ave / Essex St | M32011 | 42.362500 | -71.088220 | 42.365070 | -71.103100 | member |
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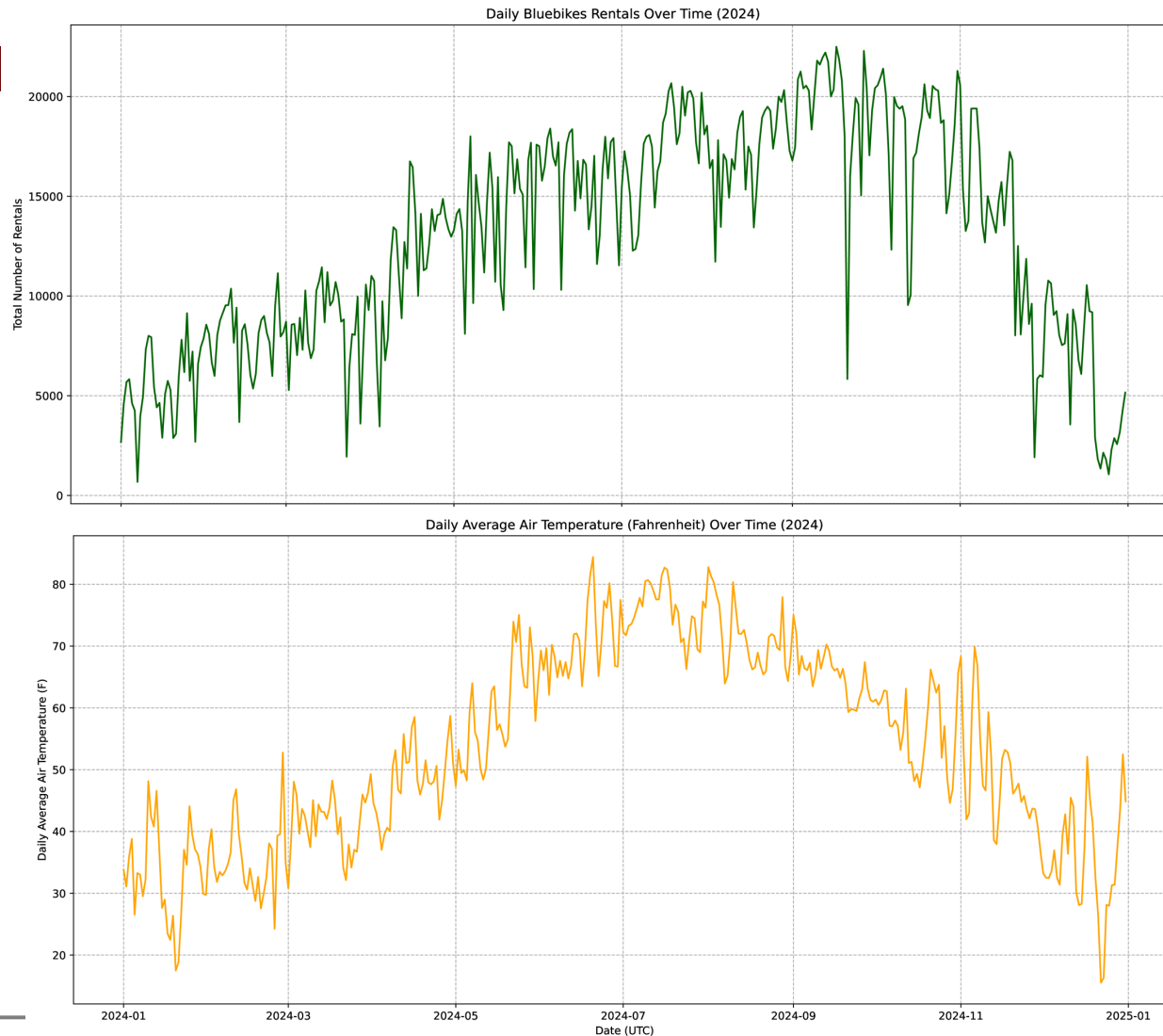
Aggregate by Day / Month / Hour ↓

| date | month | weekday | hour | rental |
|------------|-------|---------|------|--------|
| 2024-01-01 | 1 | 1 | 0 | 185.0 |
| 2024-01-01 | 1 | 1 | 1 | 208.0 |
| 2024-01-01 | 1 | 1 | 2 | 92.0 |
| 2024-01-01 | 1 | 1 | 3 | 32.0 |
| 2024-01-01 | 1 | 1 | 4 | 6.0 |
| 2024-01-01 | 1 | 1 | 5 | 14.0 |

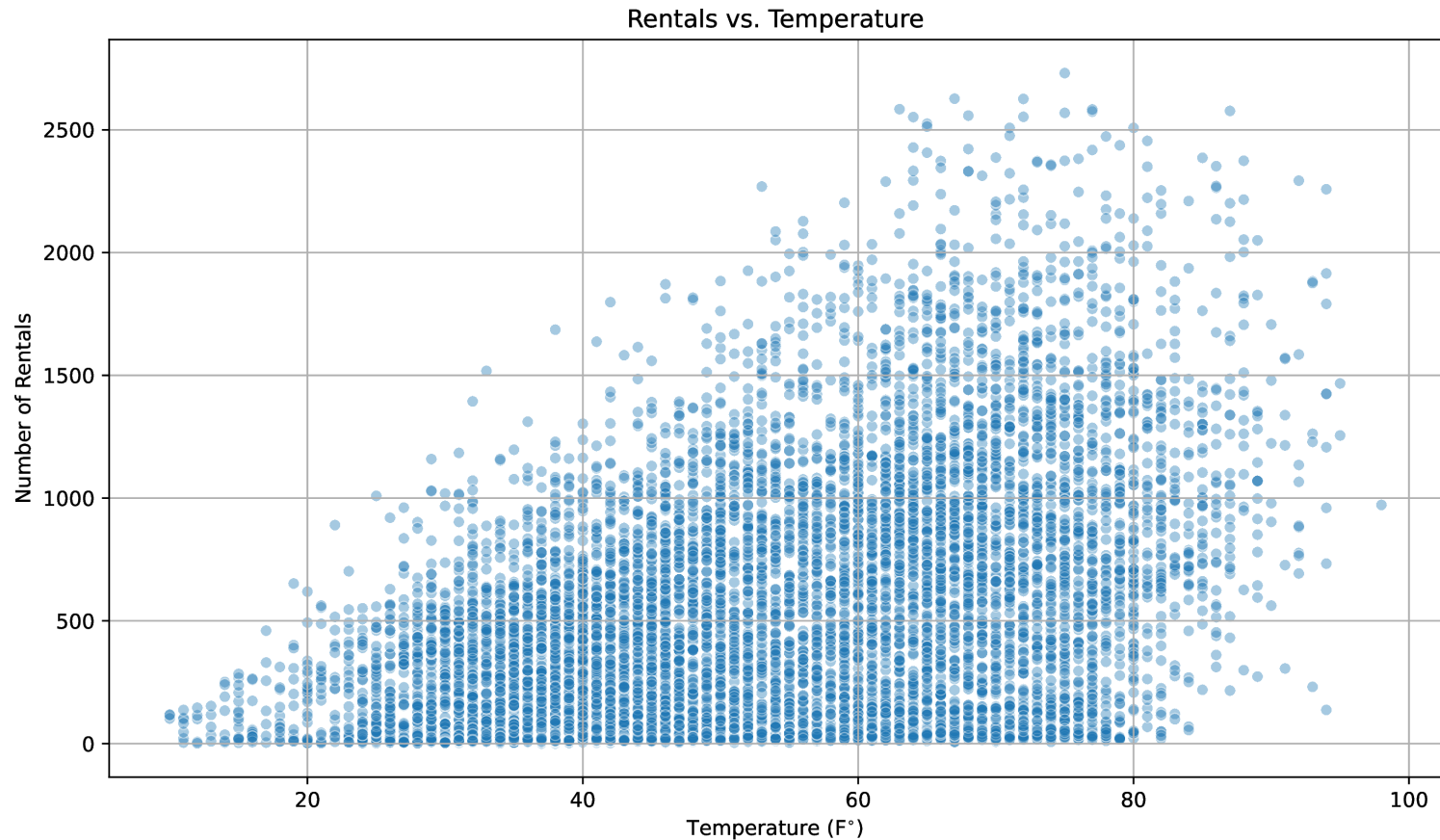
What other independent variables might we want to bring in to predict “rentals”?

What variables do *you* think affect rental volume?

Weather Data



Scatter Plot: Rentals vs Temperature



Reminder: Covariance and Correlation

For pairs of numbers $(x_1, y_1), (x_2, y_2), \dots (x_n, y_n)$, the **Covariance** (Cov) is:

$$\text{Cov}(x, y) = \frac{(x_1 - \bar{x})(y_1 - \bar{y}) + (x_2 - \bar{x})(y_2 - \bar{y}) + \dots + (x_n - \bar{x})(y_n - \bar{y})}{n}$$

$\bar{x}$ and $\bar{y}$ are averages of $x_1, \dots, x_n$ and $y_1, \dots, y_n$, respectively.

- Covariance measures the average of the product between how far x is from its mean and how far y is from its mean; Covariance can be positive, negative, or 0.
- $\text{Cov}(\text{rentals}, \text{temp}) = 3630.28$

hard to interpret...

Correlation Is More “User-Friendly”

$$\text{Cov}(\mathbf{x}, \mathbf{y}) = \frac{(\mathbf{x}_1 - \bar{\mathbf{x}})(\mathbf{y}_1 - \bar{\mathbf{y}}) + (\mathbf{x}_2 - \bar{\mathbf{x}})(\mathbf{y}_2 - \bar{\mathbf{y}}) + \cdots + (\mathbf{x}_n - \bar{\mathbf{x}})(\mathbf{y}_n - \bar{\mathbf{y}})}{n}$$

Correlation is defined as: **Correlation** is guaranteed to lie between +1 (perfect positive correlation) and -1 (perfect negative correlation).

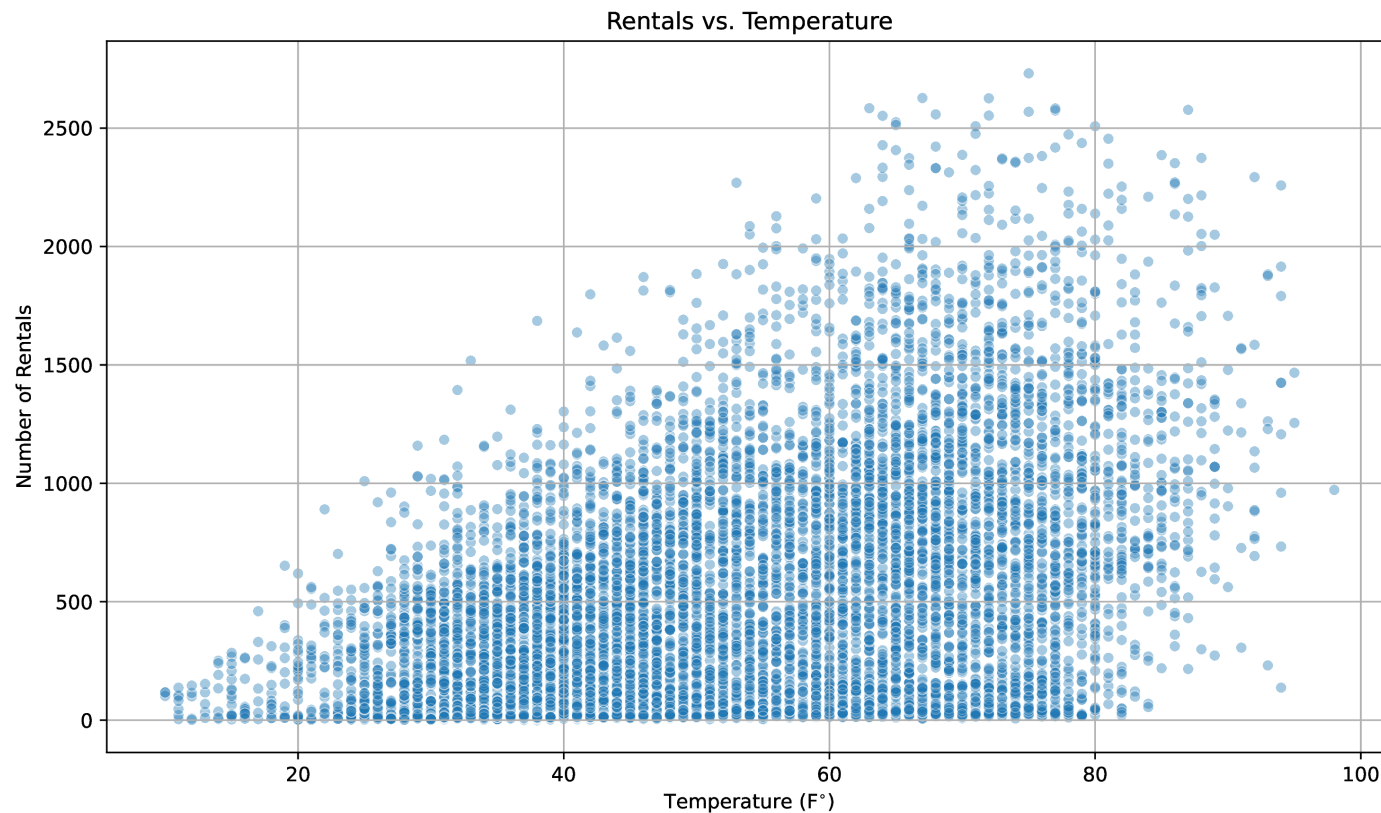
$$\text{Cor}(\mathbf{x}, \mathbf{y}) = \frac{\text{Cov}(\mathbf{x}, \mathbf{y})}{s(\mathbf{x}) \cdot s(\mathbf{y})}$$

$s(\mathbf{x})$ and $s(\mathbf{y})$: standard deviation

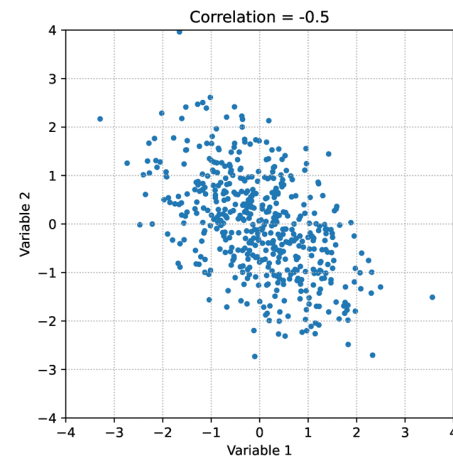
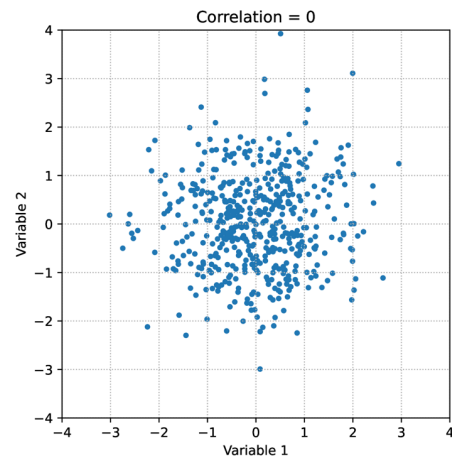
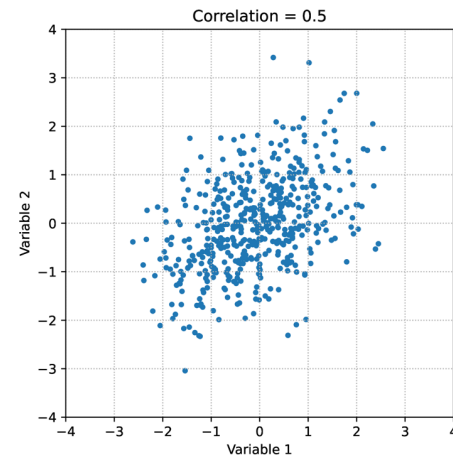
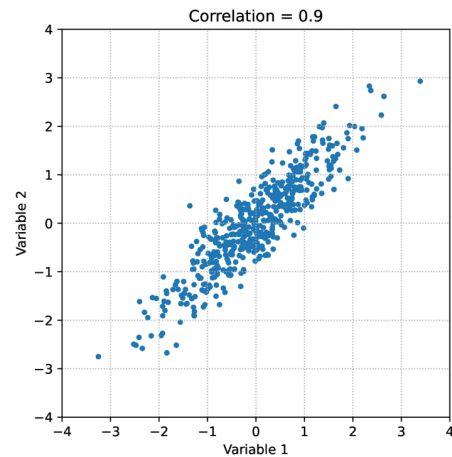
- **Covariance** is sensitive to the scale of data, whereas **correlation** is unit-less – a form of normalization
- $\text{Cor}(\text{rental}, \text{temp}) = 0.4354$

Correlation Intuition

$$\text{Cor}(\text{rentals}, \text{temp}) = 0.43$$



Correlation Intuition (cont.)



Dealing with Real Data

| date | month | weekday | hour | rental |
|------------|-------|---------|------|--------|
| 2024-01-01 | 1 | 1 | 0 | 185.0 |
| 2024-01-01 | 1 | 1 | 1 | 208.0 |
| 2024-01-01 | 1 | 1 | 2 | 92.0 |
| 2024-01-01 | 1 | 1 | 3 | 32.0 |
| 2024-01-01 | 1 | 1 | 4 | 6.0 |
| 2024-01-01 | 1 | 1 | 5 | 14.0 |

Four missing hours:

2024-01-07 04:00

2024-01-10 03:00

2024-01-18 03:00

2024-03-10 02:00

`df.describe()`

| | month | weekday | hour | rental |
|-------|-------------|-------------|-------------|-------------|
| count | 8780.000000 | 8780.000000 | 8780.000000 | 8780.000000 |
| mean | 6.515945 | 3.985763 | 11.503872 | 541.206150 |
| std | 3.450507 | 2.003392 | 6.921762 | 502.065825 |
| min | 1.000000 | 1.000000 | 0.000000 | 1.000000 |
| 25% | 4.000000 | 2.000000 | 6.000000 | 114.000000 |
| 50% | 7.000000 | 4.000000 | 12.000000 | 415.000000 |
| 75% | 10.000000 | 6.000000 | 18.000000 | 835.250000 |
| max | 12.000000 | 7.000000 | 23.000000 | 2731.000000 |

There were 8,784 hours in 2024 but we only have 8,780 rows (observations).

Why?

How to fix it?

Back from the Detour: Predicting Rentals from Time + Weather

Merge data from National Ocean and Atmospheric Administration (NOAA, Logan Airport Station) with the BlueBikes data.

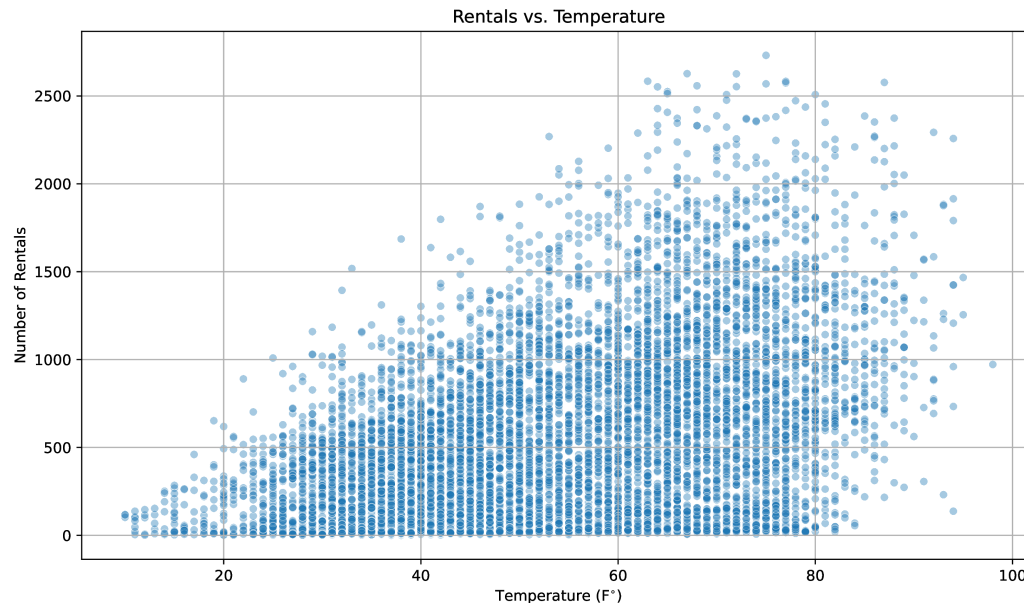
| rental | month | day | weekday | hour | temp | rel_humidity | wb_temp | precip_in | wind_speed |
|--------|-------|-----|---------|------|------|--------------|---------|-----------|------------|
| 185.0 | 1 | 1 | 1 | 0 | 34.0 | 62.5 | 23.0 | 0.0 | 9.0 |
| 208.0 | 1 | 1 | 1 | 1 | 33.0 | 63.6 | 22.0 | 0.0 | 7.0 |
| 92.0 | 1 | 1 | 1 | 2 | 33.0 | 66.3 | 23.0 | 0.0 | 7.0 |
| 32.0 | 1 | 1 | 1 | 3 | 33.0 | 66.3 | 23.0 | 0.0 | 5.0 |
| 6.0 | 1 | 1 | 1 | 4 | 33.0 | 66.3 | 23.0 | 0.0 | 3.0 |
| ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| 237.0 | 12 | 31 | 2 | 19 | 49.0 | 58.3 | 35.0 | 0.0 | 5.0 |
| 207.0 | 12 | 31 | 2 | 20 | 44.0 | 67.7 | 34.0 | 0.0 | 9.0 |
| 175.0 | 12 | 31 | 2 | 21 | 42.0 | 79.1 | 36.0 | 0.0 | 4.0 |
| 160.0 | 12 | 31 | 2 | 22 | 41.0 | 82.2 | 36.0 | 0.0 | 4.0 |
| 121.0 | 12 | 31 | 2 | 23 | 43.0 | 73.2 | 35.0 | 0.0 | 3.0 |

Revised questions:

- *“What is the number of BlueBike rentals tomorrow between 5-6 p.m. with weather predicted to be 60 degrees F, 50% humidity, and 10mph windspeed?”*
- *“How likely is it to be between 500-1000 rentals? Over 1000 rentals?”*

Predicting Rental with Temperature

Assume (for starters) that we can use one of the continuous independent variables, say “temp,” to predict hourly rentals.



“What is the number of BlueBike rentals tomorrow if temperature is 60 degrees F?”

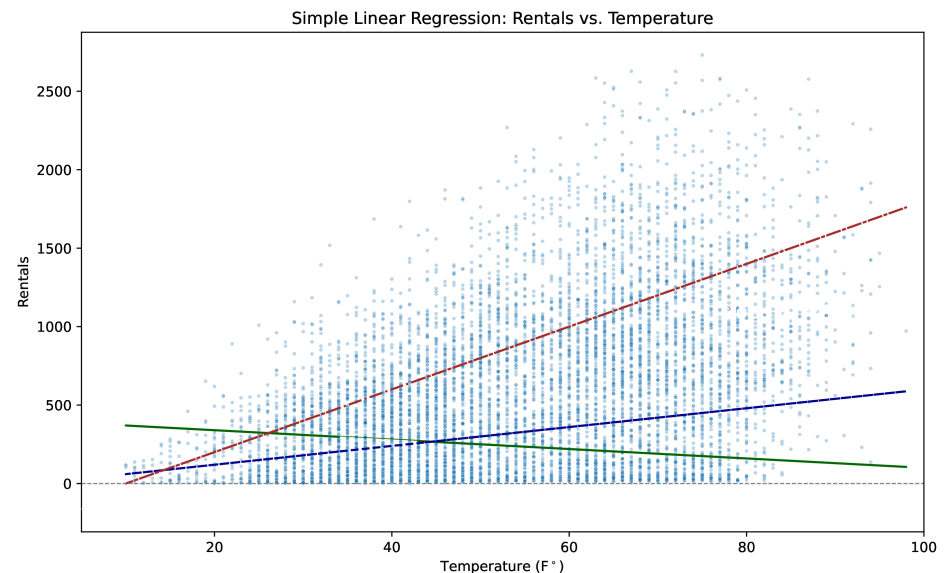
Simple Linear Regression

Predict hourly rental using temperature:

$$\hat{y} = b_0 + b_1x$$

- $\hat{y}$: dependent variable (rental)
- x : independent variable (temperature)
- b_0 : intercept
- b_1 : coefficient/slope

Different b_0 and b_1 give us different lines.



How to get one line that “best fits” the data?

Simple Linear Regression

Estimation: predict rental at temperature x_i

$$\hat{y}_i = b_0 + b_1 x_i$$

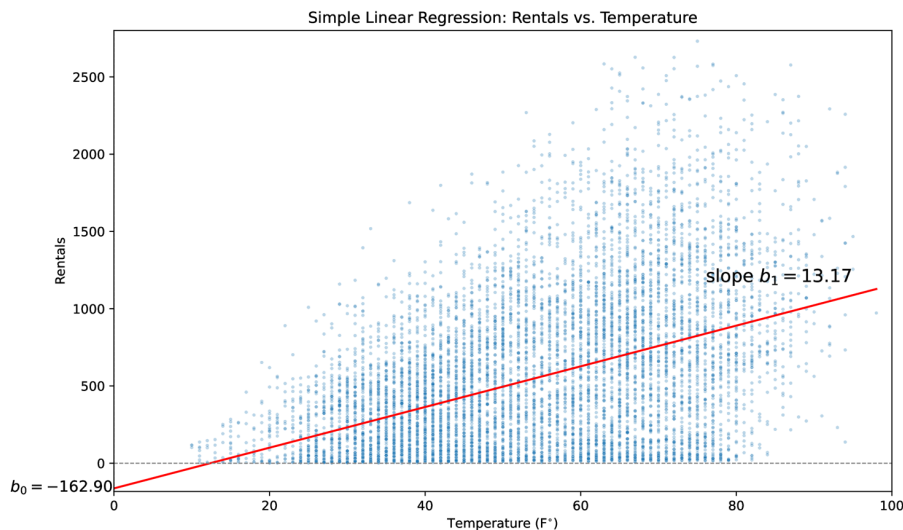
“Error” (or “Residual”) of the prediction for a point x_i :

$$e_i = y_i - \hat{y}_i$$

Look for the b_0 and b_1 values that minimize the “Sum of Squared Error” (SSE)

$$SSE = \sum_i e_i^2 = \sum (y_i - \hat{y}_i)^2$$

Find b_0 and b_1 that minimize SSE



Red line on figure minimizes SSE among all possible lines

Regression Result

```
import statsmodels.formula.api as smf
model = smf.ols('rental ~ temp', data=df).fit()
print(model.summary())
```

```

=====
                        OLS Regression Results
=====
Dep. Variable:          rental      R-squared:                0.190
Model:                  OLS        Adj. R-squared:            0.189
Method:                 Least Squares   F-statistic:              2054.
Date:                  Sat, 13 Jun 2026   Prob (F-statistic):       0.00
Time:                  17:17:43         Log-Likelihood:           -66166.
No. Observations:      8784            AIC:                     1.323e+05
Df Residuals:          8782            BIC:                     1.323e+05
Df Model:               1
Covariance Type:       nonrobust
=====

```

| | coef | std err | t | P> t | [0.025 | 0.975] |
|-----------|-----------|---------|---------|-------|----------|----------|
| Intercept | -162.9036 | 16.261 | -10.018 | 0.000 | -194.778 | -131.029 |
| temp | 13.1651 | 0.290 | 45.326 | 0.000 | 12.596 | 13.734 |

```

=====
Omnibus:                 802.574    Durbin-Watson:           0.283
Prob(Omnibus):            0.000    Jarque-Bera (JB):        1071.109
Skew:                     0.769    Prob(JB):                 2.58e-233
Kurtosis:                 3.748    Cond. No.                  189.
=====

```

Regression Result

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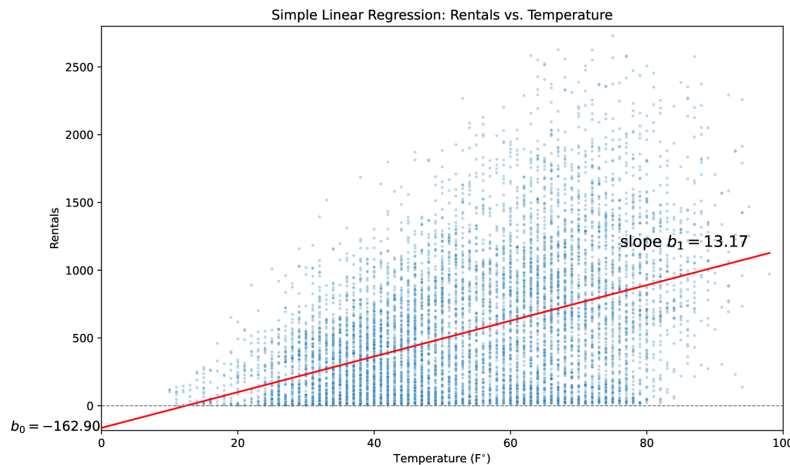
Regression Result

Values of b_0 and b_1 that yield the “best fit” line: $b_0 = -115.11$ and $b_1 = 6.02$

| OLS Regression Results | | | | | | |
|------------------------|------------------|---------------------|-----------|-------|----------|----------|
| ===== | | | | | | |
| Dep. Variable: | rental | R-squared: | 0.190 | | | |
| Model: | OLS | Adj. R-squared: | 0.189 | | | |
| Method: | Least Squares | F-statistic: | 2054. | | | |
| Date: | Sat, 13 Jun 2026 | Prob (F-statistic): | 0.00 | | | |
| Time: | 17:17:43 | Log-Likelihood: | -66166. | | | |
| No. Observations: | 8784 | AIC: | 1.323e+05 | | | |
| Df Residuals: | 8782 | BIC: | 1.323e+05 | | | |
| Df Model: | 1 | | | | | |
| Covariance Type: | nonrobust | | | | | |
| ===== | | | | | | |
| | coef | std err | t | P> t | [0.025 | 0.975] |
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| temp | 13.1651 | 0.290 | 45.326 | 0.000 | 12.596 | 13.734 |
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| Skew: | 0.769 | Prob(JB): | 2.58e-233 | | | |
| Kurtosis: | 3.748 | Cond. No. | 189. | | | |
| ===== | | | | | | |

$$\hat{y} = -162.90 + 13.17x$$

Prediction



| OLS Regression Results | | | | | | |
|------------------------|------------------|---------------------|-----------|-------|----------|----------|
| ===== | | | | | | |
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| Skew: | 0.769 | Prob(JB): | 2.58e-233 | | | |
| Kurtosis: | 3.748 | Cond. No. | 189. | | | |
| ===== | | | | | | |

$$b_0 = -162.90, b_1 = 13.17, \text{ and } \hat{y} = -162.90 + 13.17x$$

Question: What's the estimated rentals per hour if the temperature is 75 F°?

Answer: $\hat{y} = -162.90 + 13.17 \times 75 = 825$ rentals per hour

```
predicted_rental = model.predict(pd.DataFrame({'temp': [75]}))
```

R²

- Is this prediction better than the *baseline model* by always predicting $\bar{y} = 540.96$?
- $SSE = 1,794$ million < $SST = 2,214$ million
- SSE and SST are both **absolute** measures of quality of fit, whose unit does not convey much information. Let us use a **relative** measure for better comparison.

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{1,794 \text{ million}}{2,214 \text{ million}} = 0.190$$

| OLS Regression Results | | | | | | |
|------------------------|------------------|---------------------|-----------|-------|----------|----------|
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| Skew: | 0.769 | Prob(JB): | 2.58e-233 | | | |
| Kurtosis: | 3.748 | Cond. No. | 189. | | | |

From Simple to Multiple Regression

Let us add another independent variable: relative humidity (rel_humidity)

| rental | month | day | weekday | hour | temp | rel_humidity | wb_temp | precip_in | wind_speed |
|--------|-------|-----|---------|------|------|--------------|---------|-----------|------------|
| 185.0 | 1 | 1 | 1 | 0 | 34.0 | 62.5 | 23.0 | 0.0 | 9.0 |
| 208.0 | 1 | 1 | 1 | 1 | 33.0 | 63.6 | 22.0 | 0.0 | 7.0 |
| 92.0 | 1 | 1 | 1 | 2 | 33.0 | 66.3 | 23.0 | 0.0 | 7.0 |
| 32.0 | 1 | 1 | 1 | 3 | 33.0 | 66.3 | 23.0 | 0.0 | 5.0 |
| 6.0 | 1 | 1 | 1 | 4 | 33.0 | 66.3 | 23.0 | 0.0 | 3.0 |
| ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| 237.0 | 12 | 31 | 2 | 19 | 49.0 | 58.3 | 35.0 | 0.0 | 5.0 |
| 207.0 | 12 | 31 | 2 | 20 | 44.0 | 67.7 | 34.0 | 0.0 | 9.0 |
| 175.0 | 12 | 31 | 2 | 21 | 42.0 | 79.1 | 36.0 | 0.0 | 4.0 |
| 160.0 | 12 | 31 | 2 | 22 | 41.0 | 82.2 | 36.0 | 0.0 | 4.0 |
| 121.0 | 12 | 31 | 2 | 23 | 43.0 | 73.2 | 35.0 | 0.0 | 3.0 |

From Simple to Multiple Regression

```
model1 = smf.ols('rental ~ temp + rel_humidity', data=df).fit()  
print(model1.summary())
```

```
OLS Regression Results  
=====
```

| | | | |
|-------------------|------------------|---------------------|-----------|
| Dep. Variable: | rental | R-squared: | 0.220 |
| Model: | OLS | Adj. R-squared: | 0.220 |
| Method: | Least Squares | F-statistic: | 1242. |
| Date: | Sat, 13 Jun 2026 | Prob (F-statistic): | 0.00 |
| Time: | 20:59:36 | Log-Likelihood: | -65995. |
| No. Observations: | 8784 | AIC: | 1.320e+05 |
| Df Residuals: | 8781 | BIC: | 1.320e+05 |
| Df Model: | 2 | | |
| Covariance Type: | nonrobust | | |

```
=====
```

| | coef | std err | t | P> t | [0.025 | 0.975] |
|--------------|----------|---------|---------|-------|---------|---------|
| Intercept | 148.6319 | 23.100 | 6.434 | 0.000 | 103.350 | 193.914 |
| temp | 13.1160 | 0.285 | 46.037 | 0.000 | 12.557 | 13.674 |
| rel_humidity | -4.5186 | 0.242 | -18.643 | 0.000 | -4.994 | -4.044 |

```
=====
```

| | | | |
|----------------|---------|-------------------|-----------|
| Omnibus: | 814.480 | Durbin-Watson: | 0.297 |
| Prob(Omnibus): | 0.000 | Jarque-Bera (JB): | 1103.100 |
| Skew: | 0.768 | Prob(JB): | 2.92e-240 |
| Kurtosis: | 3.810 | Cond. No. | 433. |

```
=====
```

$$\text{rental} = 148 + 13.12 \times \text{temp} - 4.52 \times \text{rel_humidity}$$

If tomorrow's weather forecast is Temperature = 75F° and Humidity = 80%, how many rentals does the model predict?

$$\hat{y} = 148 + 13.12 \times 75 - 4.52 \times 80 = 770 \text{ bikes per hour}$$

What about R^2 ?

```

=====
OLS Regression Results
=====
Dep. Variable:          rental    R-squared:          0.190
Model:                  OLS      Adj. R-squared:      0.189
Method:                 Least Squares  F-statistic:        2054.
Date:                  Sat, 13 Jun 2026  Prob (F-statistic):    0.00
Time:                  17:17:43    Log-Likelihood:     -66166.
No. Observations:      8784      AIC:                1.323e+05
Df Residuals:          8782      BIC:                1.323e+05
Df Model:              1
Covariance Type:       nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
Intercept    -162.9036     16.261    -10.018     0.000    -194.778    -131.029
temp          13.1651      0.290     45.326     0.000     12.596     13.734
=====
Omnibus:            802.574    Durbin-Watson:      0.283
Prob(Omnibus):      0.000    Jarque-Bera (JB):    1071.109
Skew:               0.769    Prob(JB):            2.58e-233
Kurtosis:           3.748    Cond. No.            189.
=====

```

```

=====
OLS Regression Results
=====
Dep. Variable:          rental    R-squared:          0.220
Model:                  OLS      Adj. R-squared:      0.220
Method:                 Least Squares  F-statistic:        1242.
Date:                  Sat, 13 Jun 2026  Prob (F-statistic):    0.00
Time:                  20:59:36    Log-Likelihood:     -65995.
No. Observations:      8784      AIC:                1.320e+05
Df Residuals:          8781      BIC:                1.320e+05
Df Model:              2
Covariance Type:       nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
Intercept     148.6319     23.100      6.434     0.000     103.350     193.914
temp           13.1160      0.285     46.037     0.000     12.557     13.674
rel_humidity   -4.5186      0.242    -18.643     0.000     -4.994     -4.044
=====
Omnibus:            814.480    Durbin-Watson:      0.297
Prob(Omnibus):      0.000    Jarque-Bera (JB):    1103.100
Skew:               0.768    Prob(JB):            2.92e-240
Kurtosis:           3.810    Cond. No.            433.
=====

```

- R^2 increased from 0.19 in simple regression to 0.22 in multiple regression
- Adding an independent variable always increases R^2 :

$$R^2 = 1 - \frac{SSE}{SST}$$

Adding a new independent variable does not change SST but decreases SSE. This is because setting the coefficient of the new independent variable to zero keeps SSE unchanged, while regression tries to minimize SSE.

What about R^2 ?

- R^2 increased from 0.19 in simple regression to 0.22 in multiple regression
- Adding an independent variable always increases R^2 : $R^2 = 1 - \frac{SSE}{SST}$
- Adding a new independent variable does not change SST but decreases SSE.
- A “higher” R^2 indicates that
 - Independent variables predict the dependent variable better
- However,
 - A “high” R^2 does not guarantee error rate from the model is low enough for practical use.
 - Conversely, a model with a “low” R^2 may still be practically useful
- What constitutes a “high” or “low” R^2 is domain-dependent.

Confidence Intervals

```
model2 = smf.ols('rental ~ temp + wb_temp + rel_humidity', data=df).fit()
```

| Dep. Variable: | rental | | R-squared: | | 0.221 | |
|----------------|-----------|---------|------------|-------|----------|--------|
| | coef | std err | t | P> t | [0.025 | 0.975] |
| Intercept | -179.6712 | 110.221 | -1.630 | 0.103 | -395.730 | 36.388 |
| temp | 21.4681 | 2.757 | 7.788 | 0.000 | 16.065 | 26.872 |
| wb_temp | -8.8286 | 2.898 | -3.046 | 0.002 | -14.510 | -3.147 |
| rel_humidity | -0.7994 | 1.245 | -0.642 | 0.521 | -3.239 | 1.641 |

- Generally speaking, if a confidence interval does not contain 0, this independent variable is **significant**. Otherwise it is **insignificant**.
- Example: variable **temp** is **significant**.
- Confidence interval for **rel_humidity** contains 0, really due to **multicollinearity**.
- What if we want significance to be based on criteria other than 95% C.I.?

Multicollinearity

```
model2 = smf.ols('rental ~ temp + wb_temp + rel_humidity', data=df).fit()
```

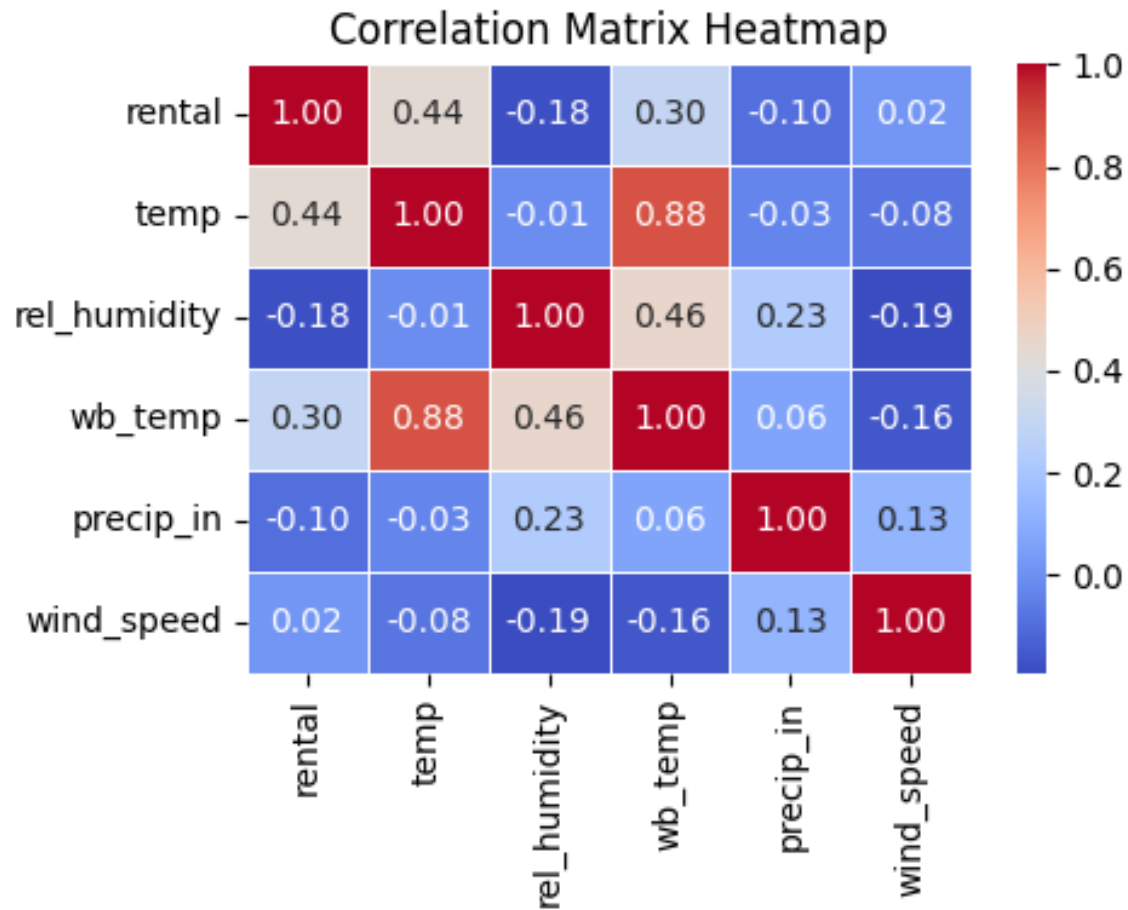
| Dep. Variable: | rental | | R-squared: | | 0.221 | |
|----------------|-----------|---------|------------|-------|----------|--------|
| | coef | std err | t | P> t | [0.025 | 0.975] |
| Intercept | -179.6712 | 110.221 | -1.630 | 0.103 | -395.730 | 36.388 |
| temp | 21.4681 | 2.757 | 7.788 | 0.000 | 16.065 | 26.872 |
| wb_temp | -8.8286 | 2.898 | -3.046 | 0.002 | -14.510 | -3.147 |
| rel_humidity | -0.7994 | 1.245 | -0.642 | 0.521 | -3.239 | 1.641 |

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.32e+03. This might indicate that there are strong **multicollinearity** or other numerical problems.

- Temperature and humidity determines wet bulb temperature!
- When this happens, coefficients are not uniquely determined.
- Rule of thumb: Do not include both variables when their correlation > 0.75.

Correlation Matrix



P Values

```
model2 = smf.ols('rental ~ temp + wb_temp + rel_humidity', data=df).fit()
```

| Dep. Variable: | rental | | R-squared: | | 0.221 | |
|----------------|-----------|---------|------------|-------|----------|--------|
| | coef | std err | t | P> t | [0.025 | 0.975] |
| Intercept | -179.6712 | 110.221 | -1.630 | 0.103 | -395.730 | 36.388 |
| temp | 21.4681 | 2.757 | 7.788 | 0.000 | 16.065 | 26.872 |
| wb_temp | -8.8286 | 2.898 | -3.046 | 0.002 | -14.510 | -3.147 |
| rel_humidity | -0.7994 | 1.245 | -0.642 | 0.521 | -3.239 | 1.641 |

- **P value:** If the coefficient is truly 0, how likely the coefficient is as different from 0 as we see purely by chance.
- The smaller the *P value*, the more confident we are that the independent variable is statistically significant.
 - Using *P value* less than 0.05 to determine significance is equivalent to the confidence interval.
 - Statistically significant variable may or may not be practically important.

Table with starts

| | Coef. | Std.Err. | t | P> t | [0.025 | 0.975] | Signif |
|--|------------|----------|----------|--------|------------|-----------|--------|
| Intercept | 90.9664 | 27.7799 | 3.2745 | 0.0011 | 36.5114 | 145.4214 | ** |
| temp | 13.1402 | 0.2854 | 46.0376 | 0.0000 | 12.5807 | 13.6997 | *** |
| rel_humidity | -4.0696 | 0.2554 | -15.9368 | 0.0000 | -4.5702 | -3.5691 | *** |
| precip_in | -1125.3808 | 212.1237 | -5.3053 | 0.0000 | -1541.1929 | -709.5687 | *** |
| wind_speed | 3.2663 | 1.1047 | 2.9569 | 0.0031 | 1.1009 | 5.4317 | ** |
| R-squared: 0.2233 ***:p<0.001 **:p<0.01 *:p<0.05 .:p<0.1 | | | | | | | |

Other Independent Variables?

| rental | month | day | weekday | hour | temp | rel_humidity | wb_temp | precip_in | wind_speed |
|--------|-------|-----|---------|------|------|--------------|---------|-----------|------------|
| 185.0 | 1 | 1 | 1 | 0 | 34.0 | 62.5 | 23.0 | 0.0 | 9.0 |
| 208.0 | 1 | 1 | 1 | 1 | 33.0 | 63.6 | 22.0 | 0.0 | 7.0 |
| 92.0 | 1 | 1 | 1 | 2 | 33.0 | 66.3 | 23.0 | 0.0 | 7.0 |
| 32.0 | 1 | 1 | 1 | 3 | 33.0 | 66.3 | 23.0 | 0.0 | 5.0 |
| 6.0 | 1 | 1 | 1 | 4 | 33.0 | 66.3 | 23.0 | 0.0 | 3.0 |
| ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| 237.0 | 12 | 31 | 2 | 19 | 49.0 | 58.3 | 35.0 | 0.0 | 5.0 |
| 207.0 | 12 | 31 | 2 | 20 | 44.0 | 67.7 | 34.0 | 0.0 | 9.0 |
| 175.0 | 12 | 31 | 2 | 21 | 42.0 | 79.1 | 36.0 | 0.0 | 4.0 |
| 160.0 | 12 | 31 | 2 | 22 | 41.0 | 82.2 | 36.0 | 0.0 | 4.0 |
| 121.0 | 12 | 31 | 2 | 23 | 43.0 | 73.2 | 35.0 | 0.0 | 3.0 |

Recall the original question we asked ourselves:

“What is the number of BlueBike rentals tomorrow between 5-6 p.m.?”

Let us include hour information

Including Hour

```
smf.ols('rental ~ temp + rel_humidity + hour', data=df).fit()
```

| Dep. Variable: | rental | | R-squared: | | 0.221 | |
|----------------|-----------|---------|------------|-------|----------|----------|
| | coef | std err | t | P> t | [0.025 | 0.975] |
| Intercept | -187.0446 | 23.386 | -7.998 | 0.000 | -232.887 | -141.202 |
| temp | 11.7827 | 0.268 | 43.991 | 0.000 | 11.258 | 12.308 |
| rel_humidity | -2.6261 | 0.232 | -11.339 | 0.000 | -3.080 | -2.172 |
| hour | 24.1370 | 0.659 | 36.629 | 0.000 | 22.845 | 25.429 |

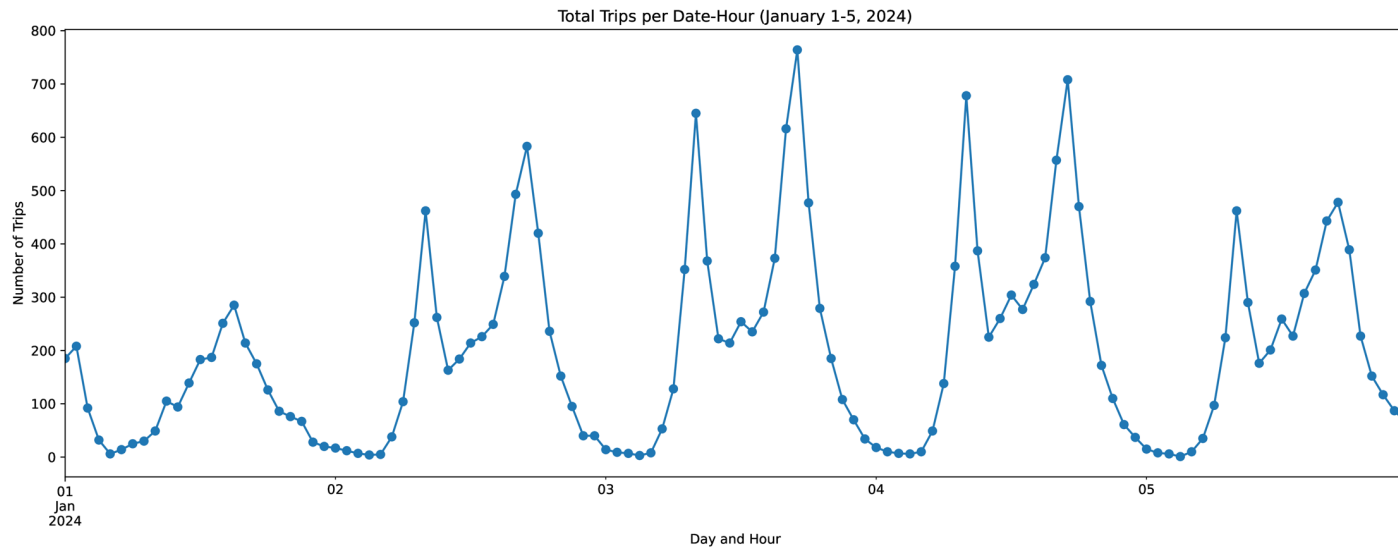
$$\text{rental} = -187.04 + 11.78 \times \text{temp} - 2.6261 \times \text{rel_humidity} + 24.14 \times \text{hour}$$

Question: Compare the hour between 11am and 12pm, with the hour between 8am and 9am. What is the model's prediction on their difference?

Answer: According to the model, 3 hours later the hourly rental increases by $24.14 \times 3 = 72.42$.

Does it make intuitive sense?

Let Us Look at Data



Daily cycle

Not linear

Categorical vs. Numerical Variables

| rental | month | day | weekday | hour | temp | rel_humidity | wb_temp | precip_in | wind_speed |
|-----------------------|-------|-----|---------|------|---------------------|--------------|---------|-----------|------------|
| 185.0 | 1 | 1 | 1 | 0 | 34.0 | 62.5 | 23.0 | 0.0 | 9.0 |
| 208.0 | 1 | 1 | 1 | 1 | 33.0 | 63.6 | 22.0 | 0.0 | 7.0 |
| 92.0 | 1 | 1 | 1 | 2 | 33.0 | 66.3 | 23.0 | 0.0 | 7.0 |
| 32.0 | 1 | 1 | 1 | 3 | 33.0 | 66.3 | 23.0 | 0.0 | 5.0 |
| 6.0 | 1 | 1 | 1 | 4 | 33.0 | 66.3 | 23.0 | 0.0 | 3.0 |
| ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| 237.0 | 12 | 31 | 2 | 19 | 49.0 | 58.3 | 35.0 | 0.0 | 5.0 |
| 207.0 | 12 | 31 | 2 | 20 | 44.0 | 67.7 | 34.0 | 0.0 | 9.0 |
| 175.0 | 12 | 31 | 2 | 21 | 42.0 | 79.1 | 36.0 | 0.0 | 4.0 |
| 160.0 | 12 | 31 | 2 | 22 | 41.0 | 82.2 | 36.0 | 0.0 | 4.0 |
| 121.0 | 12 | 31 | 2 | 23 | 43.0 | 73.2 | 35.0 | 0.0 | 3.0 |
| Categorical Variables | | | | | Numerical Variables | | | | |

Categorical vs. Numerical Variables

Numerical variables

- Value is a *number* and quantity meaningful for comparisons.
- Examples:
 - Cost
 - Temperature
 - Number of customers
 - Age

Categorical variables

- Value is a *label* that does not carry quantitative meaning.
- Examples:
 - Gender
 - Area/Zip code
 - Marital status
 - College graduate

Note: Sometimes, a categorical variable may be encoded numerically. For example:

- 1 = Monday, 2 = Tuesday, ..., 7 = Sunday
- 1 = January, 2 = February, ..., 12 = December

However, since $7 > 1$ but Sunday comes before Monday (or $12 > 1$ but December comes before January), we often treat these variables as purely categorical and ignore the numerical implications.

Treating Hour as Categorical

| | coef | std err | t | P> t | [0.025 | 0.975] |
|---------------|-----------|---------|---------|-------|----------|----------|
| Intercept | -316.5762 | 19.831 | -15.963 | 0.000 | -355.450 | -277.702 |
| C(hour)[T.1] | -23.3631 | 20.045 | -1.166 | 0.244 | -62.655 | 15.929 |
| C(hour)[T.2] | -57.6331 | 20.050 | -2.874 | 0.004 | -96.937 | -18.330 |
| C(hour)[T.3] | -82.9808 | 20.058 | -4.137 | 0.000 | -122.300 | -43.662 |
| C(hour)[T.4] | -71.9483 | 20.068 | -3.585 | 0.000 | -111.286 | -32.611 |
| C(hour)[T.5] | -9.8780 | 20.078 | -0.492 | 0.623 | -49.236 | 29.480 |
| C(hour)[T.6] | 139.2146 | 20.089 | 6.930 | 0.000 | 99.836 | 178.593 |
| C(hour)[T.7] | 451.0217 | 20.096 | 22.443 | 0.000 | 411.628 | 490.415 |
| C(hour)[T.8] | 860.4105 | 20.108 | 42.789 | 0.000 | 820.994 | 899.827 |
| C(hour)[T.9] | 608.3668 | 20.110 | 30.251 | 0.000 | 568.946 | 647.788 |
| C(hour)[T.10] | 478.7049 | 20.095 | 23.822 | 0.000 | 439.314 | 518.096 |
| C(hour)[T.11] | 516.4279 | 20.070 | 25.731 | 0.000 | 477.086 | 555.770 |
| C(hour)[T.12] | 575.3052 | 20.047 | 28.698 | 0.000 | 536.008 | 614.602 |
| C(hour)[T.13] | 566.8267 | 20.042 | 28.281 | 0.000 | 527.539 | 606.114 |
| C(hour)[T.14] | 593.9586 | 20.051 | 29.622 | 0.000 | 554.653 | 633.264 |
| C(hour)[T.15] | 693.4829 | 20.068 | 34.556 | 0.000 | 654.144 | 732.821 |
| C(hour)[T.16] | 912.3823 | 20.088 | 45.419 | 0.000 | 873.005 | 951.760 |
| C(hour)[T.17] | 1161.8346 | 20.101 | 57.798 | 0.000 | 1122.431 | 1201.238 |
| C(hour)[T.18] | 903.5643 | 20.105 | 44.943 | 0.000 | 864.155 | 942.974 |
| C(hour)[T.19] | 595.0898 | 20.106 | 29.598 | 0.000 | 555.677 | 634.502 |
| C(hour)[T.20] | 375.1325 | 20.090 | 18.672 | 0.000 | 335.751 | 414.514 |
| C(hour)[T.21] | 254.2531 | 20.070 | 12.668 | 0.000 | 214.911 | 293.595 |
| C(hour)[T.22] | 168.7433 | 20.055 | 8.414 | 0.000 | 129.431 | 208.056 |
| C(hour)[T.23] | 79.7979 | 20.045 | 3.981 | 0.000 | 40.505 | 119.091 |
| temp | 11.5900 | 0.178 | 65.201 | 0.000 | 11.242 | 11.938 |
| rel_humidity | -2.4255 | 0.156 | -15.522 | 0.000 | -2.732 | -2.119 |

```
smf.ols('rental ~ temp
+ rel_humidity
+ C(hour)',
data=df).fit()
```

24 hours in a day, but only 23
variables for hour

Treat Hour as a Numerical Variable

| rental | temp | rel_humidity | hour |
|--------|------|--------------|------|
| 185 | 34 | 62.475 | 0 |
| 208 | 33 | 63.59 | 1 |
| 92 | 33 | 66.32 | 2 |
| 32 | 33 | 66.32 | 3 |
| 6 | 33 | 66.32 | 4 |
| ... | ... | ... | ... |
| 237 | 49 | 58.3 | 19 |
| 207 | 44 | 67.66 | 20 |
| 175 | 42 | 79.11 | 21 |
| 160 | 41 | 82.22 | 22 |
| 121 | 43 | 73.16 | 23 |

Change It to 0/1 “Dummy” Variables

| rental | temp | rel_humidity | hour | Hour0 | Hour1 | Hour2 | ... | Hour22 | Hour23 |
|--------|------|--------------|-----------------|-------|-------|-------|-----|--------|--------|
| 185 | 34 | 62.475 | 0 | 1 | 0 | 0 | ... | 0 | 0 |
| 208 | 33 | 63.59 | 1 | 0 | 1 | 0 | ... | 0 | 0 |
| 92 | 33 | 66.32 | 2 | 0 | 0 | 1 | ... | 0 | 0 |
| 32 | 33 | 66.32 | 3 | 0 | 0 | 0 | ... | 0 | 0 |
| 6 | 33 | 66.32 | 4 | 0 | 0 | 0 | ... | 0 | 0 |
| ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| 237 | 49 | 58.3 | 19 | 0 | 0 | 0 | ... | 0 | 0 |
| 207 | 44 | 67.66 | 20 | 0 | 0 | 0 | ... | 0 | 0 |
| 175 | 42 | 79.11 | 21 | 0 | 0 | 0 | ... | 0 | 0 |
| 160 | 41 | 82.22 | 22 | 0 | 0 | 0 | ... | 1 | 0 |
| 121 | 43 | 73.16 | 23 | 0 | 0 | 0 | ... | 0 | 1 |

Remove One of the 0/1 “Dummy” Variables

| rental | temp | rel_humidity | hour | Hour0 | Hour1 | Hour2 | ... | Hour22 | Hour23 |
|--------|------|--------------|-----------------|------------------|-------|-------|-----|--------|--------|
| 185 | 34 | 62.475 | 0 | 1 | 0 | 0 | ... | 0 | 0 |
| 208 | 33 | 63.59 | 1 | 0 | 1 | 0 | ... | 0 | 0 |
| 92 | 33 | 66.32 | 2 | 0 | 0 | 1 | ... | 0 | 0 |
| 32 | 33 | 66.32 | 3 | 0 | 0 | 0 | ... | 0 | 0 |
| 6 | 33 | 66.32 | 4 | 0 | 0 | 0 | ... | 0 | 0 |
| ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| 237 | 49 | 58.3 | 19 | 0 | 0 | 0 | ... | 0 | 0 |
| 207 | 44 | 67.66 | 20 | 0 | 0 | 0 | ... | 0 | 0 |
| 175 | 42 | 79.11 | 21 | 0 | 0 | 0 | ... | 0 | 0 |
| 160 | 41 | 82.22 | 22 | 0 | 0 | 0 | ... | 1 | 0 |
| 121 | 43 | 73.16 | 23 | 0 | 0 | 0 | ... | 0 | 1 |

Why do we need to remove one column?

Otherwise we run into multicollinearity!

Final Data Set to Run Regression

| rental | temp | rel_humidity | Hour1 | Hour2 | ... | Hour22 | Hour23 |
|--------|------|--------------|-------|-------|-----|--------|--------|
| 185 | 34 | 62.475 | 0 | 0 | ... | 0 | 0 |
| 208 | 33 | 63.59 | 1 | 0 | ... | 0 | 0 |
| 92 | 33 | 66.32 | 0 | 1 | ... | 0 | 0 |
| 32 | 33 | 66.32 | 0 | 0 | ... | 0 | 0 |
| 6 | 33 | 66.32 | 0 | 0 | ... | 0 | 0 |
| ... | ... | ... | ... | ... | ... | ... | ... |
| 237 | 49 | 58.3 | 0 | 0 | ... | 0 | 0 |
| 207 | 44 | 67.66 | 0 | 0 | ... | 0 | 0 |
| 175 | 42 | 79.11 | 0 | 0 | ... | 0 | 0 |
| 160 | 41 | 82.22 | 0 | 0 | ... | 1 | 0 |
| 121 | 43 | 73.16 | 0 | 0 | ... | 0 | 1 |

How to Interpret the Coefficients?

| | coef | std err | t | P> t | [0.025 | 0.975] |
|---------------|-----------|---------|---------|-------|----------|----------|
| Intercept | -316.5762 | 19.831 | -15.963 | 0.000 | -355.450 | -277.702 |
| C(hour)[T.1] | -23.3631 | 20.045 | -1.166 | 0.244 | -62.655 | 15.929 |
| C(hour)[T.2] | -57.6331 | 20.050 | -2.874 | 0.004 | -96.937 | -18.330 |
| C(hour)[T.3] | -82.9808 | 20.058 | -4.137 | 0.000 | -122.300 | -43.662 |
| C(hour)[T.4] | -71.9483 | 20.068 | -3.585 | 0.000 | -111.286 | -32.611 |
| C(hour)[T.5] | -9.8780 | 20.078 | -0.492 | 0.623 | -49.236 | 29.480 |
| C(hour)[T.6] | 139.2146 | 20.089 | 6.930 | 0.000 | 99.836 | 178.593 |
| C(hour)[T.7] | 451.0217 | 20.096 | 22.443 | 0.000 | 411.628 | 490.415 |
| C(hour)[T.8] | 860.4105 | 20.108 | 42.789 | 0.000 | 820.994 | 899.827 |
| C(hour)[T.9] | 608.3668 | 20.110 | 30.251 | 0.000 | 568.946 | 647.788 |
| C(hour)[T.10] | 478.7049 | 20.095 | 23.822 | 0.000 | 439.314 | 518.096 |
| C(hour)[T.11] | 516.4279 | 20.070 | 25.731 | 0.000 | 477.086 | 555.770 |
| C(hour)[T.12] | 575.3052 | 20.047 | 28.698 | 0.000 | 536.008 | 614.602 |
| C(hour)[T.13] | 566.8267 | 20.042 | 28.281 | 0.000 | 527.539 | 606.114 |
| C(hour)[T.14] | 593.9586 | 20.051 | 29.622 | 0.000 | 554.653 | 633.264 |
| C(hour)[T.15] | 693.4829 | 20.068 | 34.556 | 0.000 | 654.144 | 732.821 |
| C(hour)[T.16] | 912.3823 | 20.088 | 45.419 | 0.000 | 873.005 | 951.760 |
| C(hour)[T.17] | 1161.8346 | 20.101 | 57.798 | 0.000 | 1122.431 | 1201.238 |
| C(hour)[T.18] | 903.5643 | 20.105 | 44.943 | 0.000 | 864.155 | 942.974 |
| C(hour)[T.19] | 595.0898 | 20.106 | 29.598 | 0.000 | 555.677 | 634.502 |
| C(hour)[T.20] | 375.1325 | 20.090 | 18.672 | 0.000 | 335.751 | 414.514 |
| C(hour)[T.21] | 254.2531 | 20.070 | 12.668 | 0.000 | 214.911 | 293.595 |
| C(hour)[T.22] | 168.7433 | 20.055 | 8.414 | 0.000 | 129.431 | 208.056 |
| C(hour)[T.23] | 79.7979 | 20.045 | 3.981 | 0.000 | 40.505 | 119.091 |
| temp | 11.5900 | 0.178 | 65.201 | 0.000 | 11.242 | 11.938 |
| rel_humidity | -2.4255 | 0.156 | -15.522 | 0.000 | -2.732 | -2.119 |

$$\begin{aligned} \text{rentals} = & -316.6 + 11.6 \times \text{temp} \\ & -2.4 \times \text{rel_humidity} \\ & -23.4 \times \text{hour1} \\ & -57.6 \times \text{hour2} \\ & \dots \\ & +79.8 \times \text{Hour23} \end{aligned}$$

Question: If tomorrow 3am (Hour 3)'s temperature is 75F° and relative humidity 68%, what is the predicted rental for the hour?

Answer: $-316.6 + 11.6 \times 75 - 2.4 \times 68 - 82.98 \times 1 = 307.2$

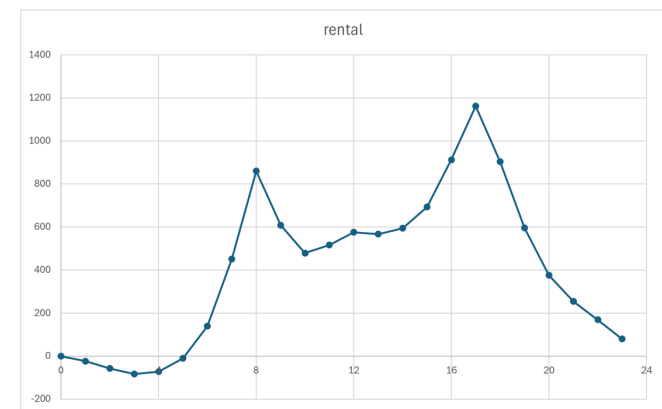
Question: What if tomorrow midnight (Hour 0)'s temperature is 75F° and relative humidity 68%?

Answer: $-316.6 + 11.6 \times 75 - 2.4 \times 68 = 390.2$

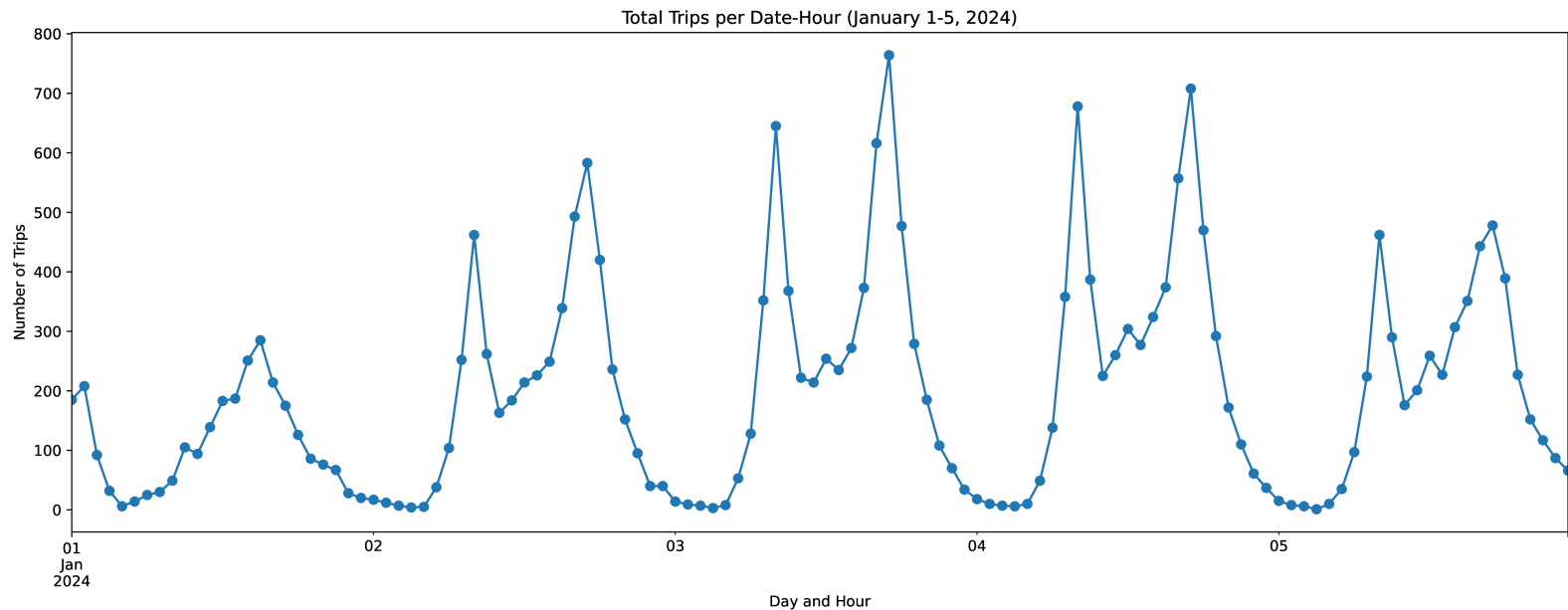
How to Interpret the Coefficients?

| | coef | std err | t | P> t | [0.025 | 0.975] |
|---------------|-----------|---------|---------|-------|----------|----------|
| Intercept | -316.5762 | 19.831 | -15.963 | 0.000 | -355.450 | -277.702 |
| C(hour)[T.1] | -23.3631 | 20.045 | -1.166 | 0.244 | -62.655 | 15.929 |
| C(hour)[T.2] | -57.6331 | 20.050 | -2.874 | 0.004 | -96.937 | -18.330 |
| C(hour)[T.3] | -82.9808 | 20.058 | -4.137 | 0.000 | -122.300 | -43.662 |
| C(hour)[T.4] | -71.9483 | 20.068 | -3.585 | 0.000 | -111.286 | -32.611 |
| C(hour)[T.5] | -9.8780 | 20.078 | -0.492 | 0.623 | -49.236 | 29.480 |
| C(hour)[T.6] | 139.2146 | 20.089 | 6.930 | 0.000 | 99.836 | 178.593 |
| C(hour)[T.7] | 451.0217 | 20.096 | 22.443 | 0.000 | 411.628 | 490.415 |
| C(hour)[T.8] | 860.4105 | 20.108 | 42.789 | 0.000 | 820.994 | 899.827 |
| C(hour)[T.9] | 608.3668 | 20.110 | 30.251 | 0.000 | 568.946 | 647.788 |
| C(hour)[T.10] | 478.7049 | 20.095 | 23.822 | 0.000 | 439.314 | 518.096 |
| C(hour)[T.11] | 516.4279 | 20.070 | 25.731 | 0.000 | 477.086 | 555.770 |
| C(hour)[T.12] | 575.3052 | 20.047 | 28.698 | 0.000 | 536.008 | 614.602 |
| C(hour)[T.13] | 566.8267 | 20.042 | 28.281 | 0.000 | 527.539 | 606.114 |
| C(hour)[T.14] | 593.9586 | 20.051 | 29.622 | 0.000 | 554.653 | 633.264 |
| C(hour)[T.15] | 693.4829 | 20.068 | 34.556 | 0.000 | 654.144 | 732.821 |
| C(hour)[T.16] | 912.3823 | 20.088 | 45.419 | 0.000 | 873.005 | 951.760 |
| C(hour)[T.17] | 1161.8346 | 20.101 | 57.798 | 0.000 | 1122.431 | 1201.238 |
| C(hour)[T.18] | 903.5643 | 20.105 | 44.943 | 0.000 | 864.155 | 942.974 |
| C(hour)[T.19] | 595.0898 | 20.106 | 29.598 | 0.000 | 555.677 | 634.502 |
| C(hour)[T.20] | 375.1325 | 20.090 | 18.672 | 0.000 | 335.751 | 414.514 |
| C(hour)[T.21] | 254.2531 | 20.070 | 12.668 | 0.000 | 214.911 | 293.595 |
| C(hour)[T.22] | 168.7433 | 20.055 | 8.414 | 0.000 | 129.431 | 208.056 |
| C(hour)[T.23] | 79.7979 | 20.045 | 3.981 | 0.000 | 40.505 | 119.091 |
| temp | 11.5900 | 0.178 | 65.201 | 0.000 | 11.242 | 11.938 |
| rel_humidity | -2.4255 | 0.156 | -15.522 | 0.000 | -2.732 | -2.119 |

- Each coefficient represents the *difference* between that hour and Hour0's rental.
- For example, the average hourly rental at 3am is lower than midnight by **82.98**.
- Which hour has the highest rental in the day? Does it make sense?



Reminder: Let Us Look at Data



Beyond Single Value: Prediction Interval

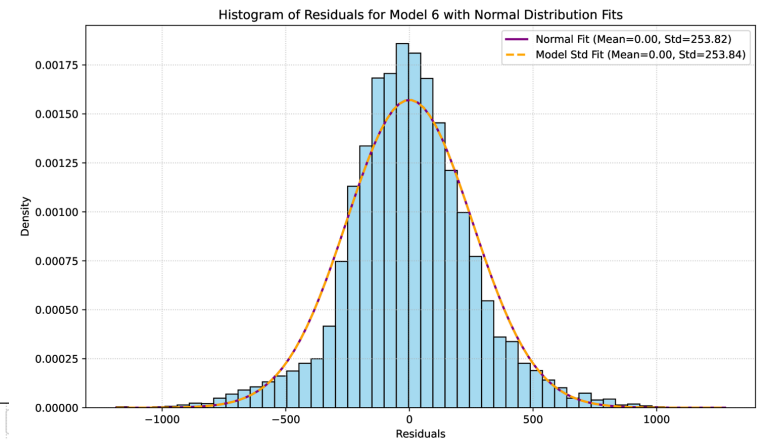
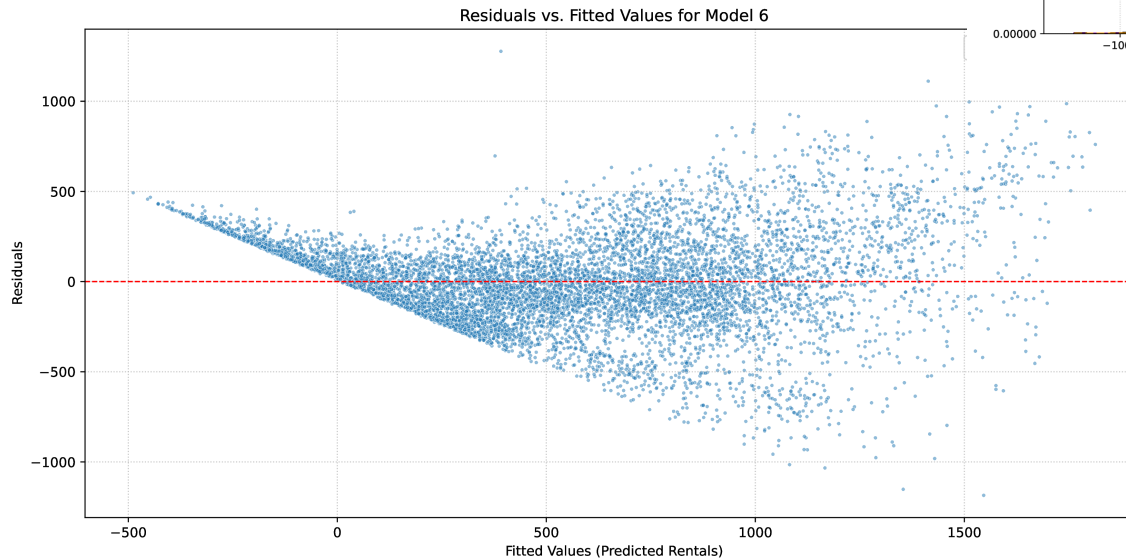
```
model6 = smf.ols('rental ~ temp + rel_humidity + precip_in + wind_speed + C(month) +  
                C(weekday) + C(hour)', data=df).fit()  
new_data = pd.DataFrame({  
    'temp': [75], 'rel_humidity': [67], 'wind_speed': [7], 'precip_in': [0],  
    'weekday': [2], 'hour': [8], 'month': [3]  
})  
prediction = model6.get_prediction(new_data).summary_frame(alpha=0.05)  
print(prediction)
```

| | mean | mean_se | mean_ci_lower | mean_ci_upper | obs_ci_lower | obs_ci_upper |
|---|-------------|-----------|---------------|---------------|--------------|--------------|
| 0 | 1100.466758 | 22.622123 | 1056.122070 | 1144.811445 | 599.671928 | 1601.261587 |

- 95% interval – $\alpha = 0.05$
- Mean Confidence Interval (1056.12,1144.81) for *mean* estimate's accuracy
- Observation Confidence Interval (or Prediction Interval) (599.67,1601.26) for predicted hourly rental value

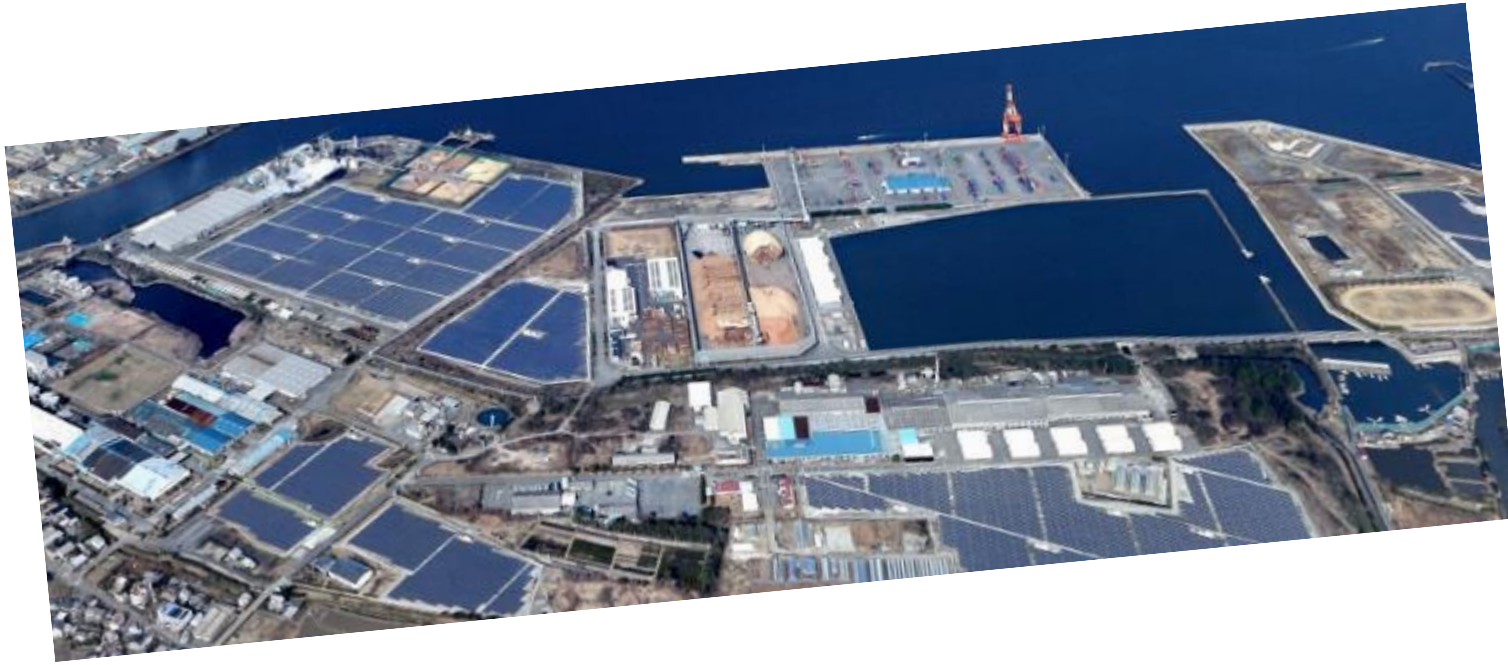
Residuals

- A better way: look at residual
 - Recall: $e_i = y_i - \hat{y}_i$



How to Select Independent Variables for a Good Model?

- Useful model
 - Easy to interpret
 - Accurate prediction *on new data*
- Potential issues to avoid:
 - Highly correlated independent variables (multicollinearity)
 - Irrelevant independent variables (high P values)
 - “Overfitting”



PREDICTING SOLAR POWER AT THE FUKUSHIMA GENERATING STATION

March 11, 2011 in Fukushima, Japan

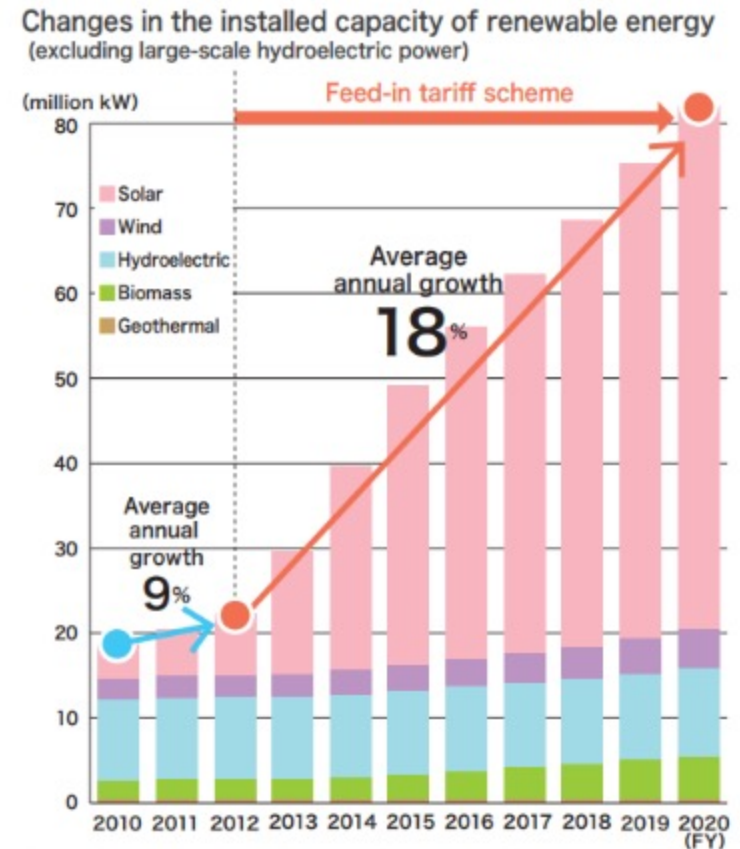


- Following a significant earthquake, a 15-meter (49 feet) tsunami disabled the cooling capability of three Fukushima Daiichi nuclear reactors, resulting in a disastrous nuclear accident
- Since then, Japan has experienced tremendous growth in renewable energy – solar, wind, geothermal, hydro, and biomass

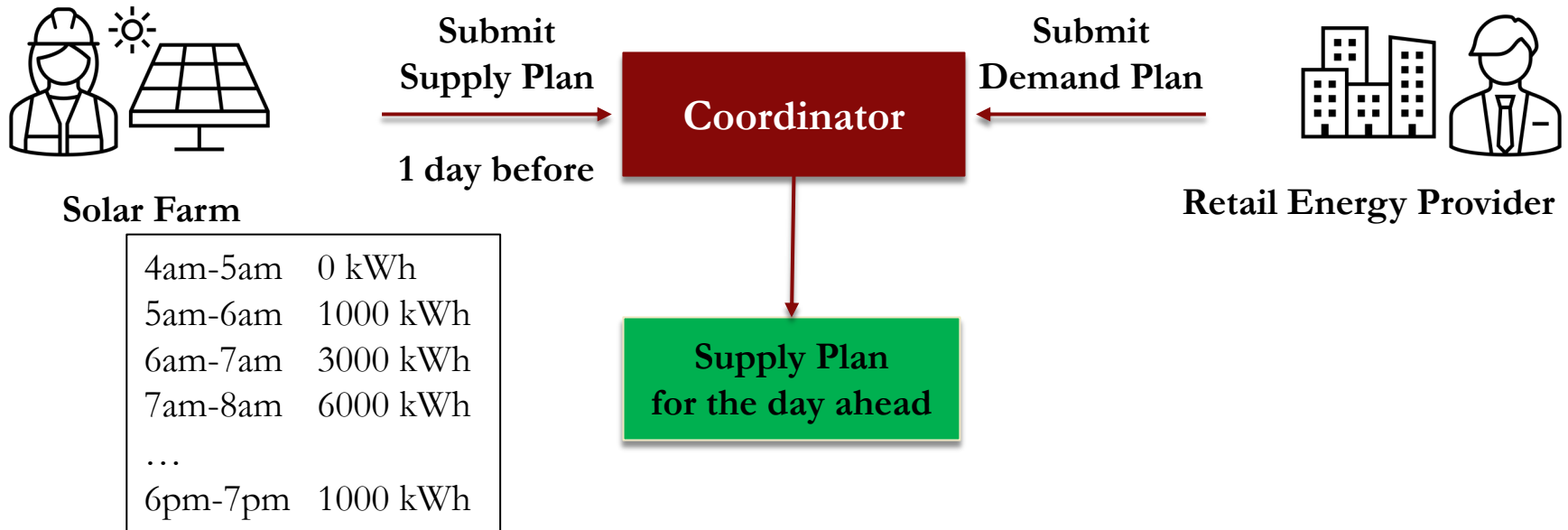
Solar Capacity Growth in Japan

Capacity Growth and Policy Impacts

- Japan's renewable capacity, especially in solar energy, surged from 2012 to 2020, significantly influenced by the **Feed-in Tariff (FIT)** scheme introduced in 2012, which offered fixed purchase prices for renewable energy.
- In 2022 FIT was replaced the **Feed-in Premium (FIP)** scheme, which combines market prices with a premium to incentivize power generators to match production with market demand and to manage supply more efficiently.
- Under FIP there is much higher cost to suppliers for mis-matched supply and demand

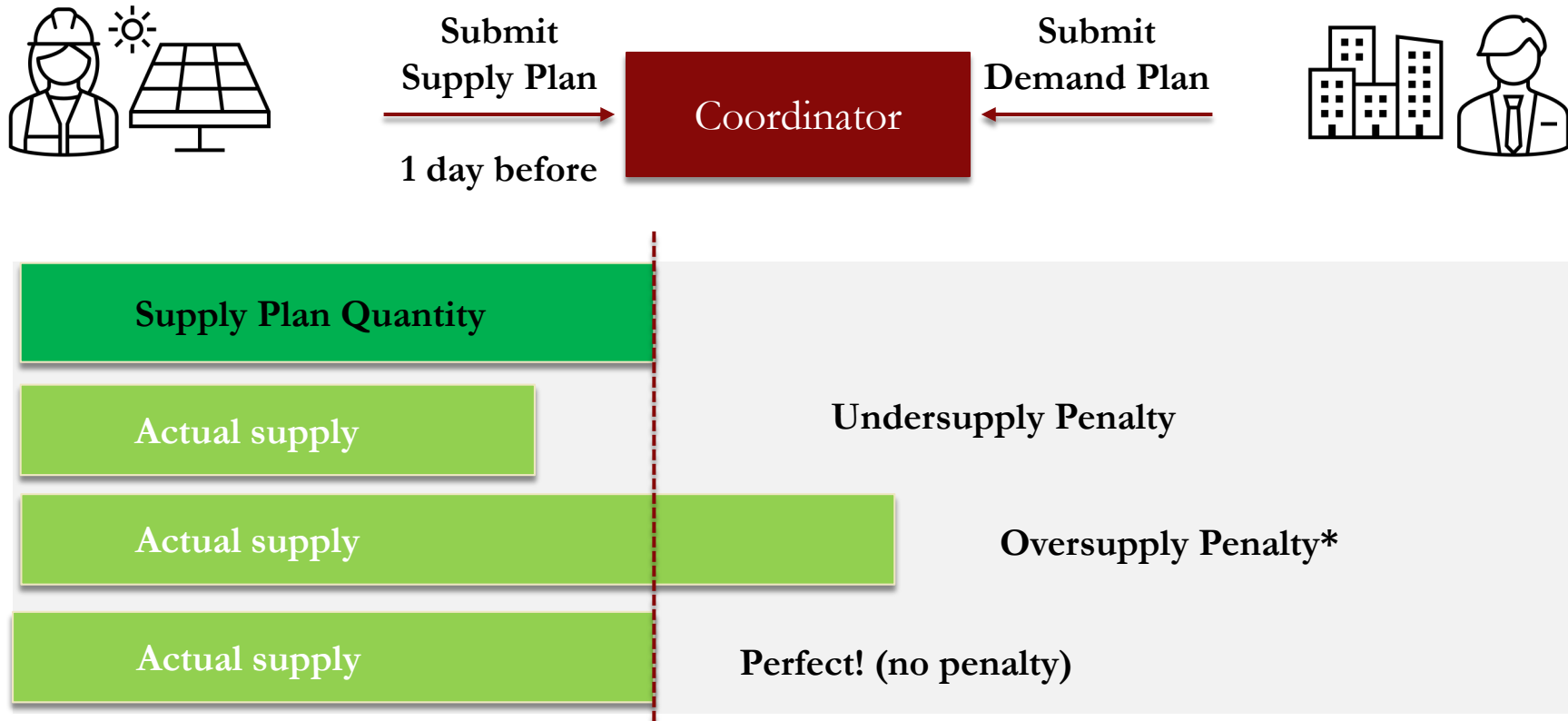


Supply-Demand Matching System



**Hour-by-Hour Production Plan
(Contract)**

Supply-Demand Mismatch Penalties



* The oversupply penalty is more of an opportunity cost penalty as they can supply the additional energy, but at a lower price than the Supply Plan price.

Example of Over/Under-Supply

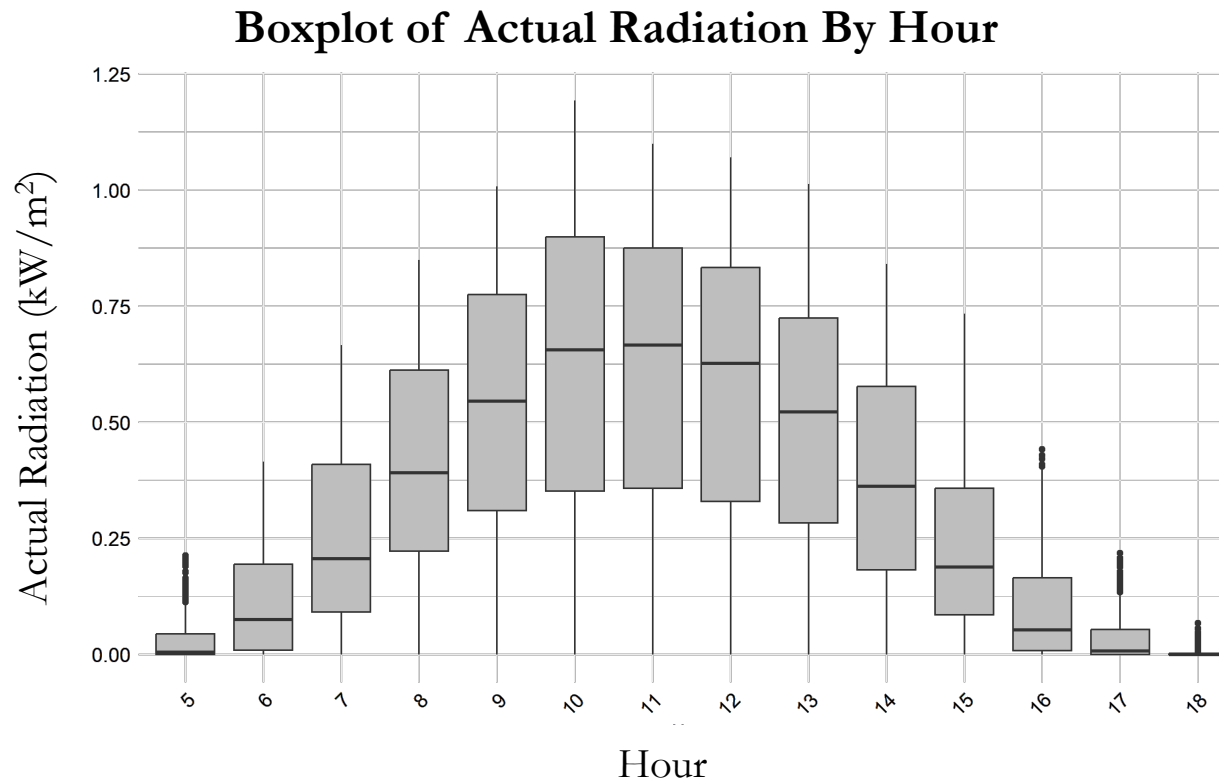
- Fukushima Generating Station (FGS) provides 6,000 kWh at 9:00-10:00 AM at a price of \$.15/kWh
- FGS internal variable production cost is \$.08/kWh
- Price for supplying more than contract demand is \$.11/kWh
- Under-supply penalty cost is \$.05/kWh
- Over-supply opportunity cost is \$.04/kWh = $0.15 - 0.11$

| | Contract “Promise” | | |
|---------------------------------|--------------------|-----------|-----------|
| | 6,000 kWh | 6,500 kWh | 5,500 kWh |
| FGS Supply | 6,000 kWh | 6,000 kWh | 6,000 kWh |
| Difference from contract | 0 kWh | -500 kWh | 500 kWh |
| Revenue | \$ 900 | \$ 900 | \$ 880 |
| Production Cost | \$ 480 | \$ 480 | \$ 480 |
| Penalty | 0 | \$ 25 | \$ 0 |
| Profit | \$ 420 | \$ 395 | \$ 400 |

- In this example, the costs of under-estimation and over-estimation are different. What are some other contexts where this can occur?

Solar generation capacity is highly variable

- By its very nature, solar power generation capacity fluctuations are large, and are influenced by time of day, day of year, air quality, cloud cover, etc.



Solar Power Data for Fukushima Generating Station (FGS)

Solar Power and Meteorological Data

k : # of independent variables ($k = 8$); n : # of observations ($n = 8461$)

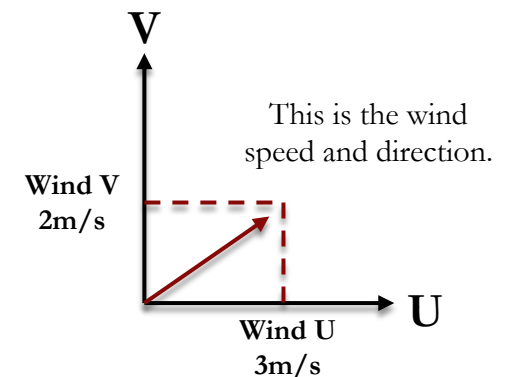


| | | Y | x_1 | x_2 | x_3 | x_4 | x_5 | x_6 | x_7 | x_8 |
|------|-----------------|------------|---------------|---------------|----------|--------|--------|-------------|-------|------------|
| # | datetime | rad_actual | rad_forecast | temp | humidity | wind_u | wind_v | cloud_cover | month | hour |
| 1 | 2021/4/2 5:00 | y_1 0.00 | x_{11} 0.00 | x_{21} 9.81 | 74.80 | -0.92 | -4.18 | 16.70 | 4 | x_{81} 5 |
| 2 | 2021/4/2 5:30 | y_2 0.03 | x_{12} 0.00 | x_{22} 9.78 | 75.90 | -1.03 | -4.16 | 14.80 | 4 | x_{82} 5 |
| 3 | 2021/4/2 6:00 | 0.12 | 0.01 | 9.75 | 77.00 | -1.15 | -4.14 | 12.90 | 4 | 6 |
| 4 | 2021/4/2 6:30 | 0.23 | 0.03 | 10.15 | 76.00 | -1.36 | -4.19 | 28.30 | 4 | 6 |
| 5 | 2021/4/2 7:00 | 0.29 | 0.05 | 10.55 | 75.00 | -1.56 | -4.24 | 43.70 | 4 | 7 |
| 6 | 2021/4/2 7:30 | 0.40 | 0.08 | 10.88 | 73.25 | -1.75 | -4.42 | 71.85 | 4 | 7 |
| 7 | 2021/4/2 8:00 | 0.43 | 0.12 | 11.21 | 71.50 | -1.94 | -4.60 | 100.00 | 4 | 8 |
| 8 | 2021/4/2 8:30 | 0.53 | 0.16 | 11.54 | 69.50 | -2.23 | -4.44 | 100.00 | 4 | 8 |
| 9 | 2021/4/2 9:00 | 0.79 | 0.19 | 11.87 | 67.50 | -2.51 | -4.27 | 100.00 | 4 | 9 |
| 10 | 2021/4/2 9:30 | 0.84 | 0.44 | 12.04 | 66.30 | -2.68 | -3.99 | 100.00 | 4 | 9 |
| 11 | 2021/4/2 10:00 | 0.92 | 0.69 | 12.20 | 65.10 | -2.85 | -3.71 | 100.00 | 4 | 10 |
| 12 | 2021/4/2 10:30 | 0.95 | 0.71 | 12.33 | 64.40 | -2.99 | -3.39 | 100.00 | 4 | 10 |
| 13 | 2021/4/2 11:00 | 0.99 | 0.74 | 12.46 | 63.70 | -3.14 | -3.07 | 100.00 | 4 | 11 |
| 14 | 2021/4/2 11:30 | 0.99 | 0.76 | 12.60 | 63.50 | -3.07 | -2.83 | 100.00 | 4 | 11 |
| 15 | 2021/4/2 12:00 | 0.97 | 0.78 | 12.75 | 63.30 | -2.99 | -2.59 | 100.00 | 4 | 12 |
| 16 | 2021/4/2 12:30 | 0.93 | 0.78 | 12.82 | 63.15 | -2.99 | -2.35 | 100.00 | 4 | 12 |
| 17 | 2021/4/2 13:00 | 0.88 | 0.79 | 12.90 | 63.00 | -3.00 | -2.11 | 100.00 | 4 | 13 |
| 18 | 2021/4/2 13:30 | 0.80 | 0.78 | 13.15 | 61.85 | -3.17 | -1.62 | 96.55 | 4 | 13 |
| 19 | 2021/4/2 14:00 | 0.73 | 0.77 | 13.40 | 60.70 | -3.35 | -1.14 | 93.10 | 4 | 14 |
| 20 | 2021/4/2 14:30 | 0.53 | 0.76 | 13.41 | 60.25 | -3.45 | -0.69 | 96.55 | 4 | 14 |
| 21 | 2021/4/2 15:00 | 0.49 | 0.75 | 13.43 | 59.80 | -3.55 | -0.23 | 100.00 | 4 | 15 |
| ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| 8461 | 2022/1/31 18:30 | 0.00 | 0.08 | 2.54 | 55.15 | 5.50 | -3.45 | 17.75 | 1 | 18 |

Solar Power Dataset

- Let us use this data to build a model to predict the value of the dependent variable/response variable ...
 - Radiation Actual (KW/m²) (not Generation Actual (KW))
- ... as a function of the independent variables/predictors/covariates/features:
 - Radiation Forecast (KW/m²)
 - Temperature (Celsius)
 - Humidity (%)
 - Wind U (east-west) Component (m/s)
 - Wind V (north-south) Component (m/s)
 - Cloud Cover (%)
 - Month (number 1-12)
 - Hour (whole number 5-18)

Meteorological variables are forecasts generated one day in advance



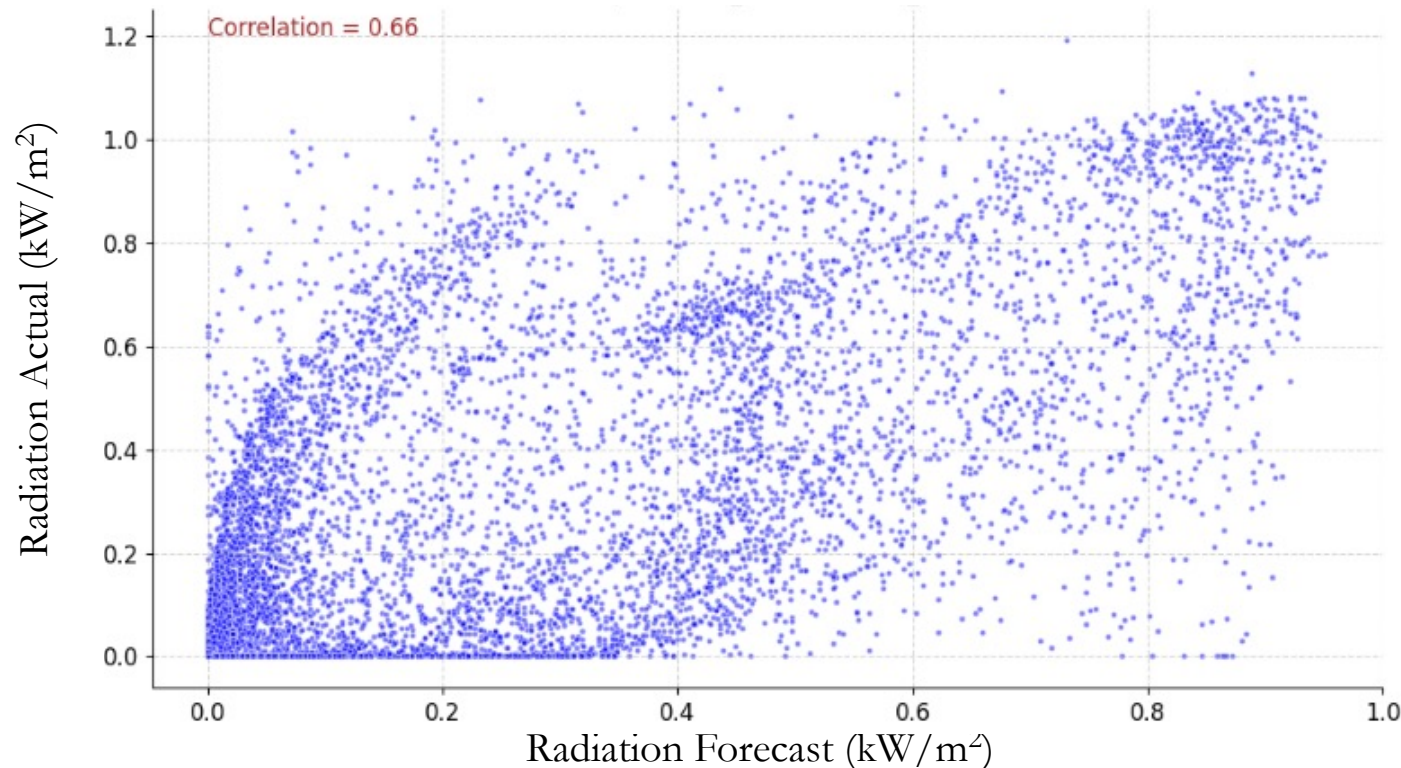
Solar Power Dataset

k : # of independent variables ($k = 8$); n : # of observations ($n = 8461$)

| | | Y | x_1 | x_2 | x_3 | x_4 | x_5 | x_6 | x_7 | x_8 | |
|------------|------|-----------------|------------|---------------|---------------|----------|--------|--------|-------------|-------|------------|
| | # | datetime | rad_actual | rad_forecast | temp | humidity | wind_u | wind_v | cloud_cover | month | hour |
| $n = 8461$ | 1 | 2021/4/2 5:00 | y_1 0.00 | x_{11} 0.00 | x_{21} 9.81 | 74.80 | -0.92 | -4.18 | 16.70 | 4 | x_{81} 5 |
| | 2 | 2021/4/2 5:30 | y_2 0.03 | x_{12} 0.00 | x_{22} 9.78 | 75.90 | -1.03 | -4.16 | 14.80 | 4 | x_{82} 5 |
| | 3 | 2021/4/2 6:00 | 0.12 | 0.01 | 9.75 | 77.00 | -1.15 | -4.14 | 12.90 | 4 | 6 |
| | 4 | 2021/4/2 6:30 | 0.23 | 0.03 | 10.15 | 76.00 | -1.36 | -4.19 | 28.30 | 4 | 6 |
| | 5 | 2021/4/2 7:00 | 0.29 | 0.05 | 10.55 | 75.00 | -1.56 | -4.24 | 43.70 | 4 | 7 |
| | 6 | 2021/4/2 7:30 | 0.40 | 0.08 | 10.88 | 73.25 | -1.75 | -4.42 | 71.85 | 4 | 7 |
| | 7 | 2021/4/2 8:00 | 0.43 | 0.12 | 11.21 | 71.50 | -1.94 | -4.60 | 100.00 | 4 | 8 |
| | 8 | 2021/4/2 8:30 | 0.53 | 0.16 | 11.54 | 69.50 | -2.23 | -4.44 | 100.00 | 4 | 8 |
| | 9 | 2021/4/2 9:00 | 0.79 | 0.19 | 11.87 | 67.50 | -2.51 | -4.27 | 100.00 | 4 | 9 |
| | 10 | 2021/4/2 9:30 | 0.84 | 0.44 | 12.04 | 66.30 | -2.68 | -3.99 | 100.00 | 4 | 9 |
| | 11 | 2021/4/2 10:00 | 0.92 | 0.69 | 12.20 | 65.10 | -2.85 | -3.71 | 100.00 | 4 | 10 |
| | 12 | 2021/4/2 10:30 | 0.95 | 0.71 | 12.33 | 64.40 | -2.99 | -3.39 | 100.00 | 4 | 10 |
| | 13 | 2021/4/2 11:00 | 0.99 | 0.74 | 12.46 | 63.70 | -3.14 | -3.07 | 100.00 | 4 | 11 |
| | 14 | 2021/4/2 11:30 | 0.99 | 0.76 | 12.60 | 63.50 | -3.07 | -2.83 | 100.00 | 4 | 11 |
| | 15 | 2021/4/2 12:00 | 0.97 | 0.78 | 12.75 | 63.30 | -2.99 | -2.59 | 100.00 | 4 | 12 |
| | 16 | 2021/4/2 12:30 | 0.93 | 0.78 | 12.82 | 63.15 | -2.99 | -2.35 | 100.00 | 4 | 12 |
| | 17 | 2021/4/2 13:00 | 0.88 | 0.79 | 12.90 | 63.00 | -3.00 | -2.11 | 100.00 | 4 | 13 |
| | 18 | 2021/4/2 13:30 | 0.80 | 0.78 | 13.15 | 61.85 | -3.17 | -1.62 | 96.55 | 4 | 13 |
| | 19 | 2021/4/2 14:00 | 0.73 | 0.77 | 13.40 | 60.70 | -3.35 | -1.14 | 93.10 | 4 | 14 |
| | 20 | 2021/4/2 14:30 | 0.53 | 0.76 | 13.41 | 60.25 | -3.45 | -0.69 | 96.55 | 4 | 14 |
| | 21 | 2021/4/2 15:00 | 0.49 | 0.75 | 13.43 | 59.80 | -3.55 | -0.23 | 100.00 | 4 | 15 |
| | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| | 8461 | 2022/1/31 18:30 | 0.00 | 0.08 | 2.54 | 55.15 | 5.50 | -3.45 | 17.75 | 1 | 18 |

How good is the FGS radiation forecast?

Radiation Forecast vs. Radiation Actual



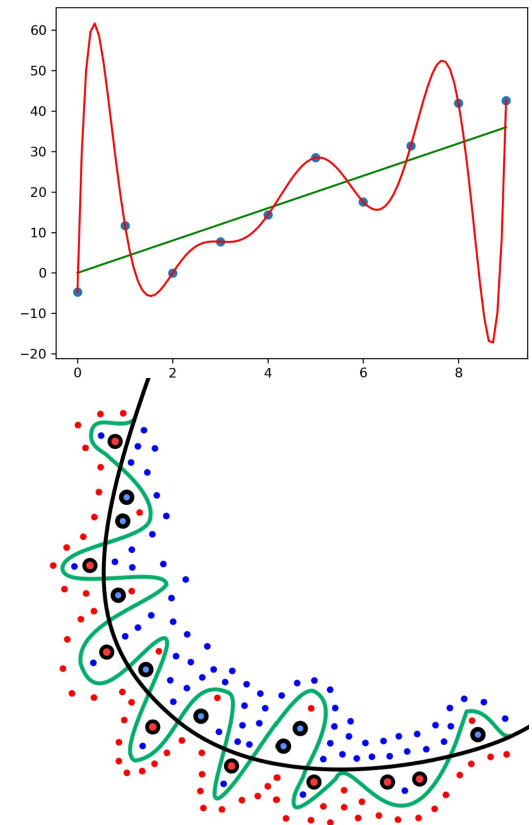
- Let us see if we can improve this forecast...

Linear Regression

- Predict the value of the dependent variable:
 - Radiation Actual (kW/m^2) at a given time (in a 30 minute increment)
- ... as a function of the independent variables:
 - Radiation Forecast (kW/m^2)
 - Temperature (Celsius)
 - Humidity (%)
 - Wind U (east-west) Component (meters/second)
 - Wind V (north-south) Component (meters/second)
 - Cloud Cover (%)
 - Month (number 1-12)
 - Hour (whole number 5-18)

Overfitting?

- Definition of Overfitting:
 - Finding patterns in the training dataset that do not generalize to the test data.
 - Idiosyncratic Noise
- Higher model complexity is both a blessing and a curse.
 - **Blessing:** Can find complex (real) patterns (e.g., non-linearities)
 - **Curse:** Prone to overfitting to noise
 - Ignorance is bliss? Simple models cannot “overthink” it.



<https://en.wikipedia.org/wiki/Overfitting>

Train/Test Split

| # | datetime | rad_actual | rad_forecast | temp | ... | hour |
|------|-----------------|------------|--------------|---------|-----|------|
| 1 | 2021/4/2 5:00 | 0.00 | 0.00 | 9.81... | | 5 |
| 2 | 2021/4/2 5:30 | 0.03 | 0.00 | 9.78... | | 5 |
| 3 | 2021/4/2 6:00 | 0.12 | 0.01 | 9.75... | | 6 |
| 4 | 2021/4/2 6:30 | 0.23 | 0.03 | 10.2... | | 6 |
| 5 | 2021/4/2 7:00 | 0.29 | 0.05 | 10.6... | | 7 |
| 6 | 2021/4/2 7:30 | 0.40 | 0.08 | 10.9... | | 7 |
| 7 | 2021/4/2 8:00 | 0.43 | 0.12 | 11.2... | | 8 |
| 8 | 2021/4/2 8:30 | 0.53 | 0.16 | 11.5... | | 8 |
| 9 | 2021/4/2 9:00 | 0.79 | 0.19 | 11.9... | | 9 |
| 10 | 2021/4/2 9:30 | 0.84 | 0.44 | 12... | | 9 |
| 11 | 2021/4/2 10:00 | 0.92 | 0.69 | 12.2... | | 10 |
| 12 | 2021/4/2 10:30 | 0.95 | 0.71 | 12.3... | | 10 |
| 13 | 2021/4/2 11:00 | 0.99 | 0.74 | 12.5... | | 11 |
| 14 | 2021/4/2 11:30 | 0.99 | 0.76 | 12.6... | | 11 |
| 15 | 2021/4/2 12:00 | 0.97 | 0.78 | 12.8... | | 12 |
| 16 | 2021/4/2 12:30 | 0.93 | 0.78 | 12.8... | | 12 |
| 17 | 2021/4/2 13:00 | 0.88 | 0.79 | 12.9... | | 13 |
| 18 | 2021/4/2 13:30 | 0.80 | 0.78 | 13.2... | | 13 |
| 19 | 2021/4/2 14:00 | 0.73 | 0.77 | 13.4... | | 14 |
| 20 | 2021/4/2 14:30 | 0.53 | 0.76 | 13.4... | | 14 |
| 21 | 2021/4/2 15:00 | 0.49 | 0.75 | 13.4... | | 15 |
| 22 | 2021/4/2 15:30 | 0.42 | 0.58 | 13.3... | | 15 |
| 23 | 2021/4/2 16:00 | 0.21 | 0.42 | 13.2... | | 16 |
| 24 | 2021/4/2 16:30 | 0.06 | 0.37 | 13... | | 16 |
| 25 | 2021/4/2 17:00 | 0.03 | 0.31 | 12.8... | | 17 |
| ... | ... | ... | ... | ... | | ... |
| 8460 | 2022/1/31 18:00 | 0.00 | 0.09 | 2.78... | | 18 |
| 8461 | 2022/1/31 18:30 | 0.00 | 0.08 | 2.54... | | 18 |

Full Dataset (n=8,461)

80% Training Set (n=6,768)

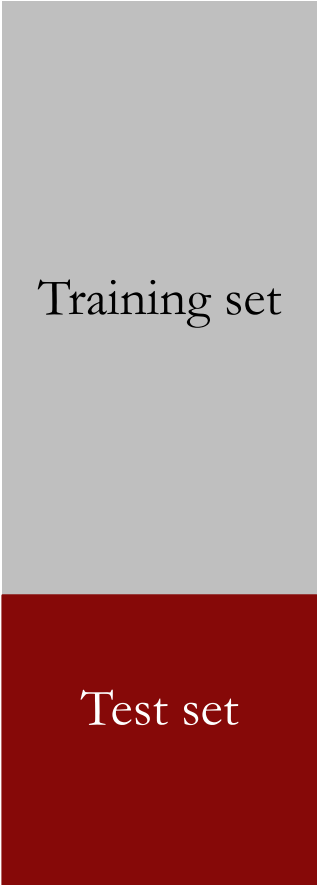
| datetime | rad_actual | rad_forecast | temp | ... | hour |
|-----------------|------------|--------------|---------|-----|------|
| 2021/4/2 5:00 | 0.00 | 0.00 | 9.81... | | 5 |
| 2021/4/2 5:30 | 0.03 | 0.00 | 9.78... | | 5 |
| 2021/4/2 6:00 | 0.12 | 0.01 | 9.75... | | 6 |
| 2021/4/2 6:30 | 0.23 | 0.03 | 10.2... | | 6 |
| 2021/4/2 7:00 | 0.29 | 0.05 | 10.6... | | 7 |
| 2021/4/2 7:30 | 0.40 | 0.08 | 10.9... | | 7 |
| 2021/4/2 8:00 | 0.43 | 0.12 | 11.2... | | 8 |
| 2021/4/2 8:30 | 0.53 | 0.16 | 11.5... | | 8 |
| 2021/4/2 9:30 | 0.84 | 0.44 | 12... | | 9 |
| 2021/4/2 10:00 | 0.92 | 0.69 | 12.2... | | 10 |
| 2021/4/2 10:30 | 0.95 | 0.71 | 12.3... | | 10 |
| 2021/4/2 11:00 | 0.99 | 0.74 | 12.5... | | 11 |
| 2021/4/2 11:30 | 0.99 | 0.76 | 12.6... | | 11 |
| 2021/4/2 12:00 | 0.97 | 0.78 | 12.8... | | 12 |
| 2021/4/2 12:30 | 0.93 | 0.78 | 12.8... | | 12 |
| 2021/4/2 13:00 | 0.88 | 0.79 | 12.9... | | 13 |
| ... | ... | ... | ... | | ... |
| 2022/1/31 18:30 | 0.00 | 0.08 | 2.54... | | 18 |

20% Testing Set (n=1,693)

| datetime | rad_actual | rad_forecast | temp | ... | hour |
|-----------------|------------|--------------|----------|-----|------|
| 2021/4/2 9:00 | 0.79 | 0.19 | 11.87... | | 9 |
| 2021/4/2 13:30 | 0.8 | 0.78 | 13.15... | | 13 |
| 2021/4/2 14:30 | 0.53 | 0.76 | 13.41... | | 14 |
| 2021/4/2 16:30 | 0.06 | 0.37 | 12.99... | | 16 |
| 2021/4/2 18:00 | 0 | 0.22 | 12.29... | | 18 |
| 2021/4/3 6:30 | 0.06 | 0.03 | 11.08... | | 6 |
| ... | ... | ... | ... | | ... |
| 2022/1/31 18:00 | 0.00 | 0.09 | 2.78... | | 18 |

Train/Test Split

- Set aside a “Test Set” of 20–50% of the data
 - Typical practice: set aside the most recently observed data: markets most recently entered, most recent observed demand, etc.
 - If there is no time-dependence, select a random Test Set
 - **Stratified sampling** is a modification that ensures similar distributions across train/test.
- Keep the Test Set “hands-off” until you validate your model
 - This procedure tests the “true” predictive power of your model



Training set

Test set

Solar Power Training Data

k : # of independent variables ($k = 8$); n : # of observations ($n = 6768$)



| datetime | rad_actual | rad_forecast | temp | humidity | wind_u | wind_v | cloud_cover | month | hour |
|-----------------|------------|--------------|-------|----------|--------|--------|-------------|-------|------|
| 2021/4/2 5:00 | 0 | 0 | 9.81 | 74.8 | -0.92 | -4.18 | 16.7 | 4 | 5 |
| 2021/4/2 5:30 | 0.03 | 0 | 9.78 | 75.9 | -1.03 | -4.16 | 14.8 | 4 | 5 |
| 2021/4/2 6:00 | 0.12 | 0.01 | 9.75 | 77 | -1.15 | -4.14 | 12.9 | 4 | 6 |
| 2021/4/2 6:30 | 0.23 | 0.03 | 10.15 | 76 | -1.36 | -4.19 | 28.3 | 4 | 6 |
| 2021/4/2 7:00 | 0.29 | 0.05 | 10.55 | 75 | -1.56 | -4.24 | 43.7 | 4 | 7 |
| 2021/4/2 7:30 | 0.4 | 0.08 | 10.88 | 73.25 | -1.75 | -4.42 | 71.85 | 4 | 7 |
| 2021/4/2 8:00 | 0.43 | 0.12 | 11.21 | 71.5 | -1.94 | -4.6 | 100 | 4 | 8 |
| 2021/4/2 8:30 | 0.53 | 0.16 | 11.54 | 69.5 | -2.23 | -4.44 | 100 | 4 | 8 |
| 2021/4/2 9:30 | 0.84 | 0.44 | 12.04 | 66.3 | -2.68 | -3.99 | 100 | 4 | 9 |
| 2021/4/2 10:00 | 0.92 | 0.69 | 12.2 | 65.1 | -2.85 | -3.71 | 100 | 4 | 10 |
| 2021/4/2 10:30 | 0.95 | 0.71 | 12.33 | 64.4 | -2.99 | -3.39 | 100 | 4 | 10 |
| 2021/4/2 11:00 | 0.99 | 0.74 | 12.46 | 63.7 | -3.14 | -3.07 | 100 | 4 | 11 |
| 2021/4/2 11:30 | 0.99 | 0.76 | 12.6 | 63.5 | -3.07 | -2.83 | 100 | 4 | 11 |
| 2021/4/2 12:00 | 0.97 | 0.78 | 12.75 | 63.3 | -2.99 | -2.59 | 100 | 4 | 12 |
| 2021/4/2 12:30 | 0.93 | 0.78 | 12.82 | 63.15 | -2.99 | -2.35 | 100 | 4 | 12 |
| 2021/4/2 13:00 | 0.88 | 0.79 | 12.9 | 63 | -3 | -2.11 | 100 | 4 | 13 |
| 2021/4/2 14:00 | 0.73 | 0.77 | 13.4 | 60.7 | -3.35 | -1.14 | 93.1 | 4 | 14 |
| 2021/4/2 15:00 | 0.49 | 0.75 | 13.43 | 59.8 | -3.55 | -0.23 | 100 | 4 | 15 |
| 2021/4/2 15:30 | 0.42 | 0.58 | 13.33 | 60.5 | -3.66 | 0.04 | 100 | 4 | 15 |
| 2021/4/2 16:00 | 0.21 | 0.42 | 13.22 | 61.2 | -3.78 | 0.32 | 100 | 4 | 16 |
| 2021/4/2 17:00 | 0.03 | 0.31 | 12.75 | 62.8 | -3.64 | 0.49 | 100 | 4 | 17 |
| 2021/4/2 17:30 | 0.01 | 0.27 | 12.52 | 64.05 | -3.46 | 0.5 | 100 | 4 | 17 |
| 2021/4/2 18:30 | 0 | 0.2 | 12.2 | 66.15 | -3.17 | 0.7 | 100 | 4 | 18 |
| 2021/4/3 5:00 | 0 | 0 | 10.89 | 70.9 | 1.77 | 0.96 | 2 | 4 | 5 |
| 2021/4/3 5:30 | 0.01 | 0 | 10.8 | 71.55 | 1.8 | 0.77 | 1 | 4 | 5 |
| ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| 2022/1/31 18:30 | 0.00 | 0.08 | 2.54 | 55.15 | 5.50 | -3.45 | 17.75 | 1 | 18 |

Model 1 with 6 Variables

- Predict the value of the dependent variable:
 - Radiation Actual (kW/m^2) at a given time (in a 30 minute increment)
- ... as a function of the independent variables:
 - Radiation Forecast (kW/m^2)
 - Temperature (Celsius)
 - Humidity (%)
 - Wind U (east-west) Component (meters/second)
 - Wind V (north-south) Component (meters/second)
 - Cloud Cover (%)
- Month (number 1-12)
- Hour (whole number 5-18)

Model 1

Regression Output of Model 1:

includes all variables except month and hour

```
model1 = smf.ols(  
    'rad_actual ~ rad_forecast + temp + humidity + wind_u +  
    wind_v + cloud_cover',  
    data = df_train)
```

1. Regression coefficient estimates

2. Indicators of the significance of the variables

```
=====
```

| | Coef. | Std.Err. | t | P> t | [0.025 | 0.975] | Signif |
|--------------|---------|----------|---------|--------|---------|---------|--------|
| Intercept | 0.2262 | 0.0190 | 11.8902 | 0.0000 | 0.1889 | 0.2635 | *** |
| rad_forecast | 0.6757 | 0.0134 | 50.2956 | 0.0000 | 0.6494 | 0.7021 | *** |
| temp | 0.0065 | 0.0006 | 10.5609 | 0.0000 | 0.0053 | 0.0078 | *** |
| humidity | -0.0028 | 0.0003 | -9.3973 | 0.0000 | -0.0034 | -0.0022 | *** |
| wind_u | 0.0033 | 0.0011 | 2.8961 | 0.0038 | 0.0011 | 0.0055 | ** |
| wind_v | -0.0053 | 0.0007 | -7.4790 | 0.0000 | -0.0067 | -0.0039 | *** |
| cloud_cover | -0.0002 | 0.0001 | -2.0723 | 0.0383 | -0.0003 | -0.0000 | * |

```
=====
```

R-squared: 0.4602

3. Indicator of model's quality of fit

Solar Power and Meteorological Data

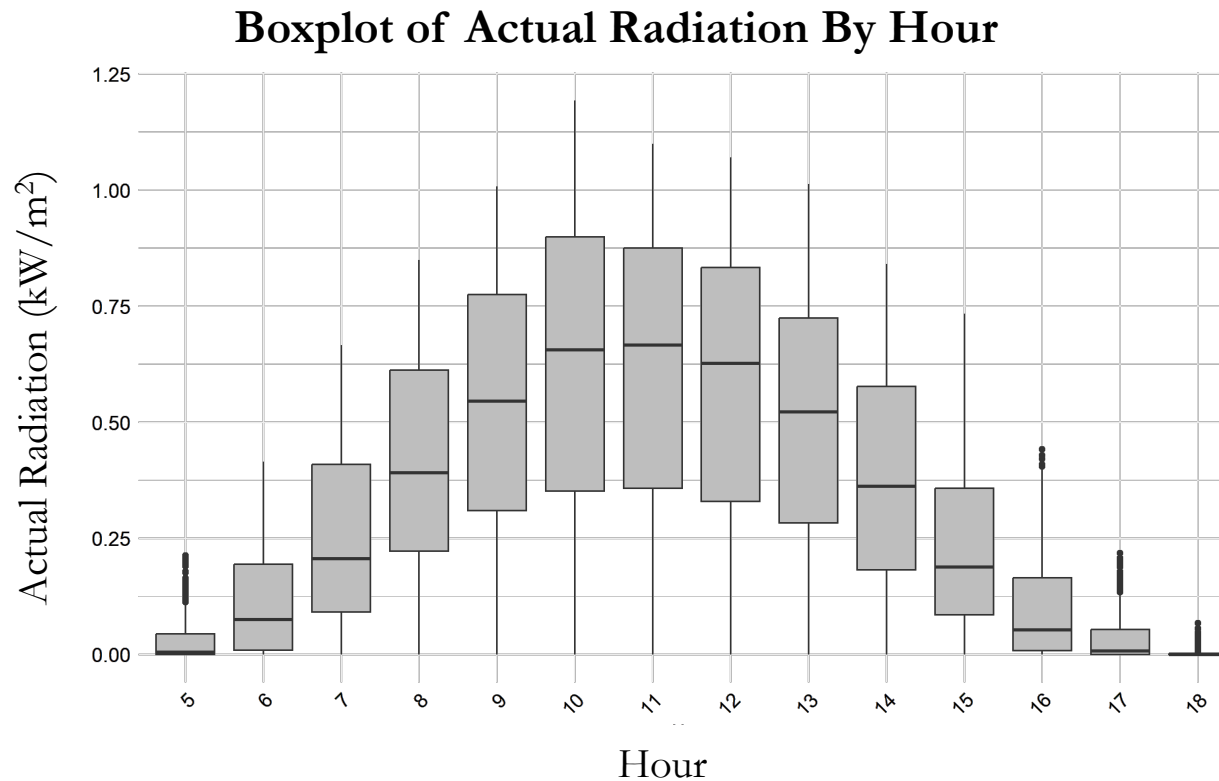
k : # of independent variables ($k = 8$); n : # of observations ($n = 8461$)



| | | Y | x_1 | x_2 | x_3 | x_4 | x_5 | x_6 | x_7 | x_8 |
|------|-----------------|------------|---------------|---------------|----------|--------|--------|-------------|-------|------------|
| # | datetime | rad_actual | rad_forecast | temp | humidity | wind_u | wind_v | cloud_cover | month | hour |
| 1 | 2021/4/2 5:00 | y_1 0.00 | x_{11} 0.00 | x_{21} 9.81 | 74.80 | -0.92 | -4.18 | 16.70 | 4 | x_{81} 5 |
| 2 | 2021/4/2 5:30 | y_2 0.03 | x_{12} 0.00 | x_{22} 9.78 | 75.90 | -1.03 | -4.16 | 14.80 | 4 | x_{82} 5 |
| 3 | 2021/4/2 6:00 | 0.12 | 0.01 | 9.75 | 77.00 | -1.15 | -4.14 | 12.90 | 4 | 6 |
| 4 | 2021/4/2 6:30 | 0.23 | 0.03 | 10.15 | 76.00 | -1.36 | -4.19 | 28.30 | 4 | 6 |
| 5 | 2021/4/2 7:00 | 0.29 | 0.05 | 10.55 | 75.00 | -1.56 | -4.24 | 43.70 | 4 | 7 |
| 6 | 2021/4/2 7:30 | 0.40 | 0.08 | 10.88 | 73.25 | -1.75 | -4.42 | 71.85 | 4 | 7 |
| 7 | 2021/4/2 8:00 | 0.43 | 0.12 | 11.21 | 71.50 | -1.94 | -4.60 | 100.00 | 4 | 8 |
| 8 | 2021/4/2 8:30 | 0.53 | 0.16 | 11.54 | 69.50 | -2.23 | -4.44 | 100.00 | 4 | 8 |
| 9 | 2021/4/2 9:00 | 0.79 | 0.19 | 11.87 | 67.50 | -2.51 | -4.27 | 100.00 | 4 | 9 |
| 10 | 2021/4/2 9:30 | 0.84 | 0.44 | 12.04 | 66.30 | -2.68 | -3.99 | 100.00 | 4 | 9 |
| 11 | 2021/4/2 10:00 | 0.92 | 0.69 | 12.20 | 65.10 | -2.85 | -3.71 | 100.00 | 4 | 10 |
| 12 | 2021/4/2 10:30 | 0.95 | 0.71 | 12.33 | 64.40 | -2.99 | -3.39 | 100.00 | 4 | 10 |
| 13 | 2021/4/2 11:00 | 0.99 | 0.74 | 12.46 | 63.70 | -3.14 | -3.07 | 100.00 | 4 | 11 |
| 14 | 2021/4/2 11:30 | 0.99 | 0.76 | 12.60 | 63.50 | -3.07 | -2.83 | 100.00 | 4 | 11 |
| 15 | 2021/4/2 12:00 | 0.97 | 0.78 | 12.75 | 63.30 | -2.99 | -2.59 | 100.00 | 4 | 12 |
| 16 | 2021/4/2 12:30 | 0.93 | 0.78 | 12.82 | 63.15 | -2.99 | -2.35 | 100.00 | 4 | 12 |
| 17 | 2021/4/2 13:00 | 0.88 | 0.79 | 12.90 | 63.00 | -3.00 | -2.11 | 100.00 | 4 | 13 |
| 18 | 2021/4/2 13:30 | 0.80 | 0.78 | 13.15 | 61.85 | -3.17 | -1.62 | 96.55 | 4 | 13 |
| 19 | 2021/4/2 14:00 | 0.73 | 0.77 | 13.40 | 60.70 | -3.35 | -1.14 | 93.10 | 4 | 14 |
| 20 | 2021/4/2 14:30 | 0.53 | 0.76 | 13.41 | 60.25 | -3.45 | -0.69 | 96.55 | 4 | 14 |
| 21 | 2021/4/2 15:00 | 0.49 | 0.75 | 13.43 | 59.80 | -3.55 | -0.23 | 100.00 | 4 | 15 |
| ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| 8461 | 2022/1/31 18:30 | 0.00 | 0.08 | 2.54 | 55.15 | 5.50 | -3.45 | 17.75 | 1 | 18 |

Solar generation capacity is highly variable

- By its very nature, radiation fluctuations are inherently powerful and vary based on factors such as time of day, date, air pollution levels, etc.



Adding Hour using categorical variables

- Add categorical variables for hour as follows:

$$\begin{aligned}\text{RadiationActual} = & \beta_0 + \beta_F \times \text{RadForecast} + \beta_T \times \text{Temp} + \beta_H \times \text{Humidity} \\ & + \beta_U \times \text{WindU} + \beta_V \times \text{WindV} + \beta_C \times \text{CloudCover} \\ & + \beta_{H6} \times \text{Hour6} + \beta_{H7} \times \text{Hour7} + \dots \\ & + \beta_{H18} \times \text{Hour18}\end{aligned}$$

- The dataset has $m = 14$ different hours, namely hours 5, 6, 7, ..., 12, ..., 18
- We need $m - 1$ “dummy” variables to work with m categories

Model results: adding Hour with categorical variables

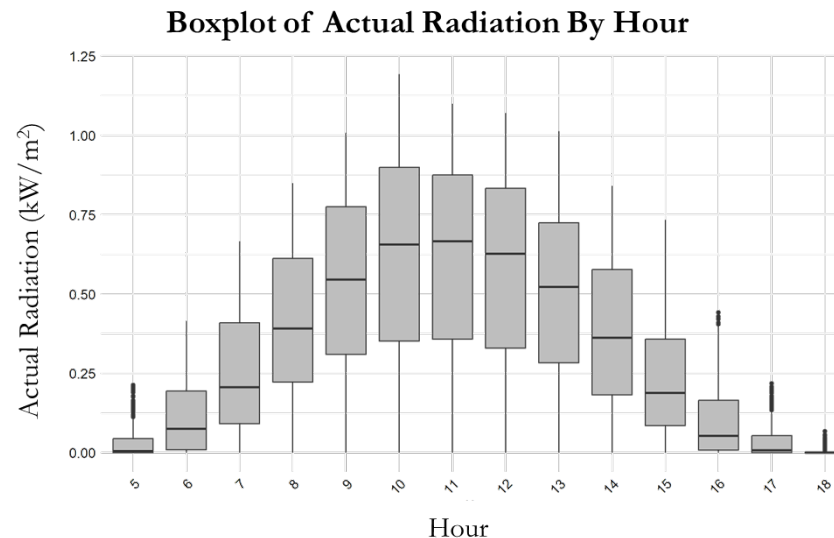
```
=====
              Coef.  Std.Err.      t  P>|t|  [0.025  0.975]  Signif
-----
Intercept      0.1150    0.0156   7.3719  0.0000   0.0844   0.1456    ***
C(hour)[T.6]   0.0720    0.0104   6.9528  0.0000   0.0517   0.0923    ***
C(hour)[T.7]   0.1932    0.0104  18.5994  0.0000   0.1729   0.2136    ***
C(hour)[T.8]   0.3097    0.0105  29.6294  0.0000   0.2892   0.3302    ***
C(hour)[T.9]   0.3737    0.0108  34.7386  0.0000   0.3526   0.3948    ***
C(hour)[T.10]  0.2947    0.0117  25.2584  0.0000   0.2719   0.3176    ***
C(hour)[T.11]  0.2769    0.0119  23.3450  0.0000   0.2537   0.3002    ***
C(hour)[T.12]  0.2302    0.0119  19.3009  0.0000   0.2068   0.2535    ***
C(hour)[T.13]  0.1668    0.0120  13.8755  0.0000   0.1432   0.1904    ***
C(hour)[T.14]  0.0462    0.0119   3.8909  0.0001   0.0229   0.0695    ***
C(hour)[T.15] -0.0556    0.0114  -4.8662  0.0000  -0.0781  -0.0332    ***
C(hour)[T.16] -0.0717    0.0107  -6.6881  0.0000  -0.0927  -0.0507    ***
C(hour)[T.17] -0.1061    0.0105 -10.0718  0.0000  -0.1268  -0.0855    ***
C(hour)[T.18] -0.1070    0.0105 -10.1982  0.0000  -0.1276  -0.0865    ***
rad_forecast    0.5648    0.0136  41.6509  0.0000   0.5382   0.5914    ***
temp           0.0067    0.0005  14.6239  0.0000   0.0058   0.0076    ***
humidity       -0.0022    0.0002 -10.1798  0.0000  -0.0026  -0.0018    ***
wind_u         0.0037    0.0008   4.5935  0.0000   0.0021   0.0053    ***
wind_v        -0.0024    0.0005  -4.6400  0.0000  -0.0034  -0.0014    ***
cloud_cover    -0.0004    0.0001  -8.0009  0.0000  -0.0005  -0.0003    ***
=====
R-squared: 0.7230
```

Model 1:
 $R^2 = 0.4602$

Model with Numerical Hour:
 $R^2 = 0.5733$

Model with Categorical Hour:
 $R^2 = 0.7230$

Changing Hour to a Categorical Variable improves the model performance



- Changing Hour to a categorical variable addresses its nonlinear relationship with solar radiation
- A linear regression model cannot otherwise capture this relationship adequately if Hour is treated as a numerical value
- Such **feature engineering** enables the model to consider the distinct impact of each hour separately, thereby improving the model's performance

Let's try some different models with Hour and Month variables

| | | Hour Variable | | |
|----------------|----------------------|----------------|------------------|----------------------|
| | | Excluded | Numeric Variable | Categorical Variable |
| Month Variable | Excluded | Model 1 | | |
| | Numeric Variable | | | |
| | Categorical Variable | | | |

Add Hour as numeric variable

| | | Hour Variable | | |
|----------------|----------------------|---------------|------------------|----------------------|
| | | Excluded | Numeric Variable | Categorical Variable |
| Month Variable | Excluded | Model 1 | ✓ | |
| | Numeric Variable | | | |
| | Categorical Variable | | | |

Add Month as numeric variable

| | | Hour Variable | | |
|----------------|----------------------|---------------|------------------|----------------------|
| | | Excluded | Numeric Variable | Categorical Variable |
| Month Variable | Excluded | Model 1 | | |
| | Numeric Variable | ✓ | | |
| | Categorical Variable | | | |

Add Month as categorical variable

| | | Hour Variable | | |
|----------------|----------------------|---------------|------------------|----------------------|
| | | Excluded | Numeric Variable | Categorical Variable |
| Month Variable | Excluded | Model 1 | | |
| | Numeric Variable | | | |
| | Categorical Variable | ✓ | | |

Add Month and Hour as categorical variables

| | | Hour Variable | | |
|----------------|----------------------|---------------|------------------|----------------------|
| | | Excluded | Numeric Variable | Categorical Variable |
| Month Variable | Excluded | Model 1 | | |
| | Numeric Variable | | | |
| | Categorical Variable | | | ✓ |

Add Month as numeric variable

| | | Hour Variable | | |
|----------------|----------------------|---------------|------------------|----------------------|
| | | Excluded | Numeric Variable | Categorical Variable |
| Month Variable | Excluded | Model 1 | | |
| | Numeric Variable | ✓ | | |
| | Categorical Variable | | | |

Model 1 + Month: Adds 'month' as a continuous variable

Model 2 ($R^2 = 0.4616$)

```

=====
              Coef.  Std.Err.      t    P>|t|  [0.025  0.975] Signif
-----+-----
Intercept      0.2545    0.0201   12.6324  0.0000    0.2150    0.2939    ***
rad_forecast    0.6630    0.0138   48.2097  0.0000    0.6360    0.6900    ***
temp           0.0072    0.0006   11.2869  0.0000    0.0060    0.0085    ***
humidity      -0.0029    0.0003   -9.6615  0.0000   -0.0034   -0.0023    ***
wind_u         0.0032    0.0011    2.7905  0.0053    0.0009    0.0054     **
wind_v        -0.0056    0.0007   -7.8088  0.0000   -0.0070   -0.0042    ***
cloud_cover    -0.0002    0.0001   -2.5022  0.0124   -0.0003   -0.0000     *
month         -0.0038    0.0009   -4.2249  0.0000   -0.0056   -0.0020    ***
=====
R-squared:  0.4616

```

Add Month as categorical variable

| | | Hour Variable | | |
|----------------|----------------------|---------------|------------------|----------------------|
| | | Excluded | Numeric Variable | Categorical Variable |
| Month Variable | Excluded | Model 1 | | |
| | Numeric Variable | | | |
| | Categorical Variable | ✓ | | |

Model 3 ($R^2 = 0.469$)

Model 1 + Month: Adds 'month' as a categorical variable

| | Coef. | Std.Err. | t | P> t | [0.025 | 0.975] | Signif |
|-------------------|---------|----------|----------|--------|---------|---------|--------|
| Intercept | 0.2890 | 0.0225 | 12.8476 | 0.0000 | 0.2449 | 0.3330 | *** |
| C(month)[T.4] | 0.0420 | 0.0156 | 2.6860 | 0.0072 | 0.0113 | 0.0726 | ** |
| C(month)[T.5] | 0.0739 | 0.0188 | 3.9222 | 0.0001 | 0.0369 | 0.1108 | *** |
| C(month)[T.6] | 0.1318 | 0.0216 | 6.1108 | 0.0000 | 0.0895 | 0.1740 | *** |
| C(month)[T.7] | 0.0724 | 0.0259 | 2.7975 | 0.0052 | 0.0217 | 0.1231 | ** |
| C(month)[T.8] | 0.0620 | 0.0271 | 2.2860 | 0.0223 | 0.0088 | 0.1152 | * |
| C(month)[T.9] | 0.0311 | 0.0235 | 1.3211 | 0.1865 | -0.0150 | 0.0771 | |
| C(month)[T.10] | 0.0302 | 0.0199 | 1.5192 | 0.1288 | -0.0088 | 0.0692 | |
| C(month)[T.11] | 0.0056 | 0.0164 | 0.3390 | 0.7346 | -0.0266 | 0.0377 | |
| C(month)[T.12] | 0.0069 | 0.0129 | 0.5325 | 0.5944 | -0.0184 | 0.0321 | |
| rad_forecast | 0.6409 | 0.0144 | 44.6552 | 0.0000 | 0.6127 | 0.6690 | *** |
| temp | 0.0050 | 0.0013 | 3.9843 | 0.0001 | 0.0026 | 0.0075 | *** |
| humidity | -0.0038 | 0.0003 | -11.7940 | 0.0000 | -0.0044 | -0.0031 | *** |
| wind_u | 0.0028 | 0.0012 | 2.3875 | 0.0170 | 0.0005 | 0.0051 | * |
| wind_v | -0.0059 | 0.0007 | -7.9136 | 0.0000 | -0.0074 | -0.0045 | *** |
| cloud_cover | -0.0002 | 0.0001 | -2.7806 | 0.0054 | -0.0004 | -0.0001 | ** |
| R-squared: 0.4690 | | | | | | | |

Add Hour as numeric variable

| | | Hour Variable | | |
|----------------|----------------------|---------------|------------------|----------------------|
| | | Excluded | Numeric Variable | Categorical Variable |
| Month Variable | Excluded | Model 1 | ✓ | |
| | Numeric Variable | | | |
| | Categorical Variable | | | |

Model 4 ($R^2 = 0.5733$)

Model 1 + Hour: Adds 'hour' as a continuous variable

| | Coef. | Std.Err. | t | P> t | [0.025 | 0.975] | Signif |
|-------------------|---------|----------|----------|--------|---------|---------|--------|
| Intercept | 0.5114 | 0.0182 | 28.0869 | 0.0000 | 0.4757 | 0.5471 | *** |
| rad_forecast | 0.7898 | 0.0122 | 64.5013 | 0.0000 | 0.7658 | 0.8139 | *** |
| temp | 0.0061 | 0.0006 | 11.1535 | 0.0000 | 0.0051 | 0.0072 | *** |
| humidity | -0.0029 | 0.0003 | -11.1027 | 0.0000 | -0.0034 | -0.0024 | *** |
| wind_u | 0.0027 | 0.0010 | 2.6733 | 0.0075 | 0.0007 | 0.0047 | ** |
| wind_v | -0.0040 | 0.0006 | -6.3760 | 0.0000 | -0.0053 | -0.0028 | *** |
| cloud_cover | 0.0000 | 0.0001 | 0.3021 | 0.7626 | -0.0001 | 0.0001 | |
| hour | -0.0269 | 0.0006 | -42.3427 | 0.0000 | -0.0282 | -0.0257 | *** |
| R-squared: 0.5733 | | | | | | | |

Add Hour as categorical variable

| | | Hour Variable | |
|----------------|----------------------|---------------|----------------------|
| | | Excluded | Categorical Variable |
| Month Variable | Excluded | Model 1 | ✓ |
| | Numeric Variable | | |
| | Categorical Variable | | |

Model 5 ($R^2 = 0.723$)

Model 1 + Hour: Adds 'hour' as a categorical variable

| | Coef. | Std.Err. | t | P> t | [0.025 | 0.975] | Signif |
|-------------------|---------|----------|----------|--------|---------|---------|--------|
| Intercept | 0.1150 | 0.0156 | 7.3719 | 0.0000 | 0.0844 | 0.1456 | *** |
| C(hour)[T.6] | 0.0720 | 0.0104 | 6.9528 | 0.0000 | 0.0517 | 0.0923 | *** |
| C(hour)[T.7] | 0.1932 | 0.0104 | 18.5994 | 0.0000 | 0.1729 | 0.2136 | *** |
| C(hour)[T.8] | 0.3097 | 0.0105 | 29.6294 | 0.0000 | 0.2892 | 0.3302 | *** |
| C(hour)[T.9] | 0.3737 | 0.0108 | 34.7386 | 0.0000 | 0.3526 | 0.3948 | *** |
| C(hour)[T.10] | 0.2947 | 0.0117 | 25.2584 | 0.0000 | 0.2719 | 0.3176 | *** |
| C(hour)[T.11] | 0.2769 | 0.0119 | 23.3450 | 0.0000 | 0.2537 | 0.3002 | *** |
| C(hour)[T.12] | 0.2302 | 0.0119 | 19.3009 | 0.0000 | 0.2068 | 0.2535 | *** |
| C(hour)[T.13] | 0.1668 | 0.0120 | 13.8755 | 0.0000 | 0.1432 | 0.1904 | *** |
| C(hour)[T.14] | 0.0462 | 0.0119 | 3.8909 | 0.0001 | 0.0229 | 0.0695 | *** |
| C(hour)[T.15] | -0.0556 | 0.0114 | -4.8662 | 0.0000 | -0.0781 | -0.0332 | *** |
| C(hour)[T.16] | -0.0717 | 0.0107 | -6.6881 | 0.0000 | -0.0927 | -0.0507 | *** |
| C(hour)[T.17] | -0.1061 | 0.0105 | -10.0718 | 0.0000 | -0.1268 | -0.0855 | *** |
| C(hour)[T.18] | -0.1070 | 0.0105 | -10.1982 | 0.0000 | -0.1276 | -0.0865 | *** |
| rad_forecast | 0.5648 | 0.0136 | 41.6509 | 0.0000 | 0.5382 | 0.5914 | *** |
| temp | 0.0067 | 0.0005 | 14.6239 | 0.0000 | 0.0058 | 0.0076 | *** |
| humidity | -0.0022 | 0.0002 | -10.1798 | 0.0000 | -0.0026 | -0.0018 | *** |
| wind_u | 0.0037 | 0.0008 | 4.5935 | 0.0000 | 0.0021 | 0.0053 | *** |
| wind_v | -0.0024 | 0.0005 | -4.6400 | 0.0000 | -0.0034 | -0.0014 | *** |
| cloud_cover | -0.0004 | 0.0001 | -8.0009 | 0.0000 | -0.0005 | -0.0003 | *** |
| R-squared: 0.7230 | | | | | | | |

Add Month and Hour as categorical variables

| | | Hour Variable | |
|----------------|----------------------|----------------|----------------------|
| | | Excluded | Categorical Variable |
| Month Variable | Excluded | Model 1 | |
| | Numeric Variable | | |
| | Categorical Variable | | ✓ |

Model 6 ($R^2 = 0.735$)

Model 1 + Month, Hour: Adds 'month' & 'hour' as a categorical variable

| | Coef. | Std.Err. | t | P> t | [0.025 | 0.975] | Signif |
|----------------|---------|----------|----------|--------|---------|---------|--------|
| Intercept | 0.1744 | 0.0175 | 9.9573 | 0.0000 | 0.1401 | 0.2088 | *** |
| C(month)[T.4] | 0.0566 | 0.0113 | 5.0336 | 0.0000 | 0.0346 | 0.0787 | *** |
| C(month)[T.5] | 0.0663 | 0.0136 | 4.8852 | 0.0000 | 0.0397 | 0.0930 | *** |
| C(month)[T.6] | 0.1236 | 0.0156 | 7.9376 | 0.0000 | 0.0930 | 0.1541 | *** |
| C(month)[T.7] | 0.0538 | 0.0186 | 2.8950 | 0.0038 | 0.0174 | 0.0903 | ** |
| C(month)[T.8] | 0.0295 | 0.0194 | 1.5196 | 0.1287 | -0.0086 | 0.0676 | |
| C(month)[T.9] | 0.0105 | 0.0168 | 0.6245 | 0.5323 | -0.0224 | 0.0434 | |
| C(month)[T.10] | 0.0010 | 0.0142 | 0.0711 | 0.9433 | -0.0267 | 0.0287 | |
| C(month)[T.11] | -0.0195 | 0.0117 | -1.6701 | 0.0949 | -0.0423 | 0.0034 | . |
| C(month)[T.12] | -0.0110 | 0.0091 | -1.2076 | 0.2273 | -0.0289 | 0.0069 | |
| C(hour)[T.6] | 0.0729 | 0.0101 | 7.1894 | 0.0000 | 0.0530 | 0.0928 | *** |
| C(hour)[T.7] | 0.1961 | 0.0102 | 19.2607 | 0.0000 | 0.1761 | 0.2160 | *** |
| C(hour)[T.8] | 0.3132 | 0.0103 | 30.5122 | 0.0000 | 0.2930 | 0.3333 | *** |
| C(hour)[T.9] | 0.3873 | 0.0106 | 36.4807 | 0.0000 | 0.3665 | 0.4082 | *** |
| C(hour)[T.10] | 0.3384 | 0.0118 | 28.7314 | 0.0000 | 0.3153 | 0.3615 | *** |
| C(hour)[T.11] | 0.3210 | 0.0120 | 26.7814 | 0.0000 | 0.2975 | 0.3445 | *** |
| C(hour)[T.12] | 0.2767 | 0.0121 | 22.8872 | 0.0000 | 0.2530 | 0.3004 | *** |
| C(hour)[T.13] | 0.2135 | 0.0122 | 17.5139 | 0.0000 | 0.1896 | 0.2374 | *** |
| C(hour)[T.14] | 0.0912 | 0.0120 | 7.5895 | 0.0000 | 0.0677 | 0.1148 | *** |
| C(hour)[T.15] | -0.0166 | 0.0115 | -1.4402 | 0.1499 | -0.0391 | 0.0060 | |
| C(hour)[T.16] | -0.0527 | 0.0106 | -4.9615 | 0.0000 | -0.0736 | -0.0319 | *** |
| C(hour)[T.17] | -0.0909 | 0.0104 | -8.7270 | 0.0000 | -0.1113 | -0.0705 | *** |
| C(hour)[T.18] | -0.0961 | 0.0103 | -9.2974 | 0.0000 | -0.1164 | -0.0759 | *** |
| rad_forecast | 0.4583 | 0.0149 | 30.7157 | 0.0000 | 0.4290 | 0.4875 | *** |
| temp | 0.0070 | 0.0009 | 7.7269 | 0.0000 | 0.0052 | 0.0087 | *** |
| humidity | -0.0033 | 0.0002 | -14.2912 | 0.0000 | -0.0038 | -0.0029 | *** |
| wind_u | 0.0035 | 0.0008 | 4.2354 | 0.0000 | 0.0019 | 0.0051 | *** |
| wind_v | -0.0032 | 0.0005 | -6.0630 | 0.0000 | -0.0043 | -0.0022 | *** |
| cloud_cover | -0.0006 | 0.0001 | -11.0703 | 0.0000 | -0.0007 | -0.0005 | *** |

R-squared: 0.7350

R^2 values of the different models

| | | Hour Variable | | |
|----------------|----------------------|---------------|------------------|----------------------|
| | | Excluded | Numeric Variable | Categorical Variable |
| Month Variable | Excluded | .4602 | .5733 | .7230 |
| | Numeric Variable | .4616 | .5736 | |
| | Categorical Variable | .4690 | | .7350 |

- The inclusion of additional variables increases the R^2 value
- Including Hour as a categorical variable markedly enhances the overall R^2 value

Out-of-Sample Testing

- Here is our model 1, based on the train data (n=6,768) :
$$\text{Radiation Actual} = 0.2262 + 0.6757 \times \text{Rad Forecast} + 0.006541 \times \text{Temp} \\ - 0.002779 \times \text{Humidity} + 0.003281 \times \text{Wind U} - 0.005324 \times \text{Wind V} \\ - 0.0001515 \times \text{Cloud Cover}$$
- Let us use the model to compute predictions and residuals for each observation in the Test Set (n=1,693)
 - For example, here is our prediction for May 11, 2021, at 10:30:
$$\text{Prediction} = 0.2262 + 0.6757 \times (0.7286) + 0.006541 \times (17.80) \dots = 0.6907$$

$$\text{Actual Radiation} = 0.9274$$

$$\text{Residual } (e_i) = 0.9274 - 0.6907 = 0.2367$$
- How good is this prediction? Well, let's look at all of the Test Set data and compute *prediction metrics*

Recall R^2

- By how much does the regression model improve the quality of the (in-sample) prediction, as compared to the baseline model?
- Percent reduction in **SSE** over **SST**.

| Actual Radiation | Model Prediction | Baseline (Average) |
|------------------|------------------|--------------------|
| y_1 | $\hat{y}_1$ | $\bar{y}$ |
| y_2 | $\hat{y}_2$ | $\bar{y}$ |
| ... | ... | |
| y_n | $\hat{y}_n$ | $\bar{y}$ |

$$R^2 = 1 - \frac{\text{Sum of squared errors of regression model (SSE)}}{\text{Sum of squared errors of baseline model (SST)}}$$
$$= 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where $\bar{y} = \frac{1}{N} \cdot \sum_{i=1}^N y_i$

Out-of-Sample R^2 (OSR²)

- Out-of-Sample R^2 : by how much does the regression model improve the quality of the (out-of-sample) predictions, as compared to the baseline model?

$$\begin{aligned} OSR^2 &= 1 - \frac{\text{Sum of squared residuals of regression model on the Test Set}}{\text{Sum of squared residuals of baseline model on the Test Set}} \\ &= 1 - \frac{\sum_{i \in \text{Test Set}} (y_i - \hat{y}_i)^2}{\sum_{i \in \text{Test Set}} (y_i - \bar{y})^2} \end{aligned} \quad \bar{y} = \text{Average in the } \underline{\text{Training}} \text{ set}$$

- In general, it is better for the OSR² to be close to the in-sample R^2
 - Otherwise, it may be indicative of intrinsic differences between the training set and the test set, or of the model being “overfit” to the training data
- We prefer higher OSR²

In-Sample R^2 vs. Out-of-Sample R^2

80% Training Set (n=6,768)

| Time/Date | rad_actual | | Predictions | Baseline |
|-----------------|------------|-----|-------------|----------|
| 4/2/2021 5:00 | 0.00 | ... | 0.02 | 0.316 |
| 4/2/2021 5:30 | 0.03 | ... | 0.12 | 0.316 |
| 4/2/2021 6:00 | 0.12 | ... | 0.15 | 0.316 |
| 4/2/2021 6:30 | 0.23 | ... | 0.33 | 0.316 |
| ... | ... | ... | | ... |
| 1/31/2022 18:30 | 0.00 | ... | 0.07 | 0.316 |

$$R^2 = 1 - \frac{\sum_{i \in \text{Train Set}} (y_i - \hat{y}_i)^2}{\sum_{i \in \text{Train Set}} (y_i - \bar{y})^2}$$

Both baselines use the average of the training set $\bar{y} = 0.316 \text{ kW/m}^2$

20% Testing Set (n=1,693)

| Time/Date | rad_actual | | Predictions | Baseline |
|----------------|------------|-----|-------------|----------|
| 11/28/2021 1pm | 0.65 | ... | 0.51 | 0.316 |
| 11/27/2021 4pm | 0.01 | ... | 0.09 | 0.316 |
| ... | ... | ... | ... | ... |
| 8/23/2022 9:30 | 0.36 | ... | 0.40 | 0.316 |

$$OSR^2 = 1 - \frac{\sum_{i \in \text{Test Set}} (y_i - \hat{y}_i)^2}{\sum_{i \in \text{Test Set}} (y_i - \bar{y})^2}$$

Computing OSR² in Python

- You can read the In-Sample R² right from the model summary, but OSR² requires calculation.

```
predictions = model.predict(df_test)

ssr = np.sum((y_test_actual - predictions)**2)

sst = np.sum((y_test_actual - np.mean(y_train_actual))**2)

OSR2 = 1 - ssr / sst
```

Out-of-Sample Assessment of all Models

| R ² | | | | | OSR ² | | | | |
|----------------|----------------------|---------------|------------------|----------------------|------------------|----------------------|---------------|------------------|----------------------|
| | | Hour Variable | | | | | Hour Variable | | |
| | | Excluded | Numeric Variable | Categorical Variable | | | Excluded | Numeric Variable | Categorical Variable |
| Month Variable | Excluded | .46 | .57 | .72 | Month Variable | Excluded | .45 | .56 | .72 |
| | Numeric Variable | .46 | .57 | | | Numeric Variable | .45 | .56 | |
| | Categorical Variable | .47 | | .73 | | Categorical Variable | .45 | | .73 |

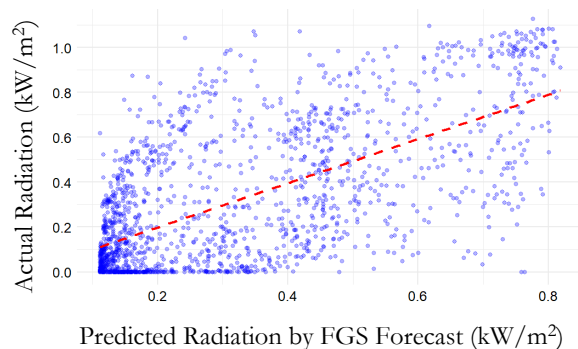
- By and large, the OSR² is very consistent with the in-sample R²
- If OSR² is much smaller than the R², what does this usually mean?

Visualizing the Predictions of the Models

- Scatter plots of predicted versus actual values for different models

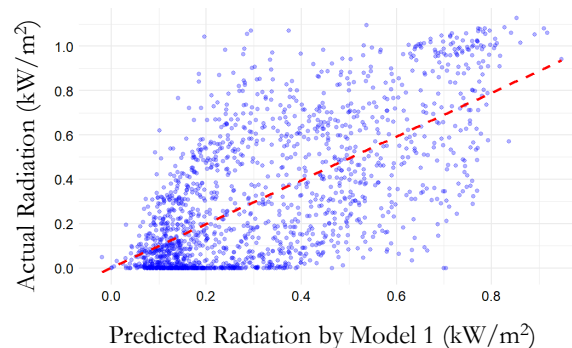
FGS Forecast

(Use only radiation forecast)



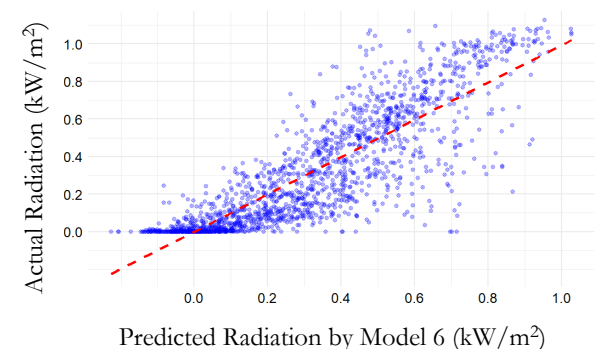
$$R^2 = 0.44$$
$$OSR^2 = 0.43$$

Model 1



$$R^2 = 0.46$$
$$OSR^2 = 0.45$$

Model 6



$$R^2 = 0.73$$
$$OSR^2 = 0.73$$

Value Impact of the Models

Estimating the value of using predictions from Model 6

- Assume FGS contracts for their forecast
- Contract price is \$.15/kWh
- FGS internal variable production cost is \$.08/kWh
- Price for supplying more than contract demand is \$.11/kWh
- Under-supply penalty cost is \$.05/kWh
- Implement on the Test Set hours
- Extrapolate to annual values

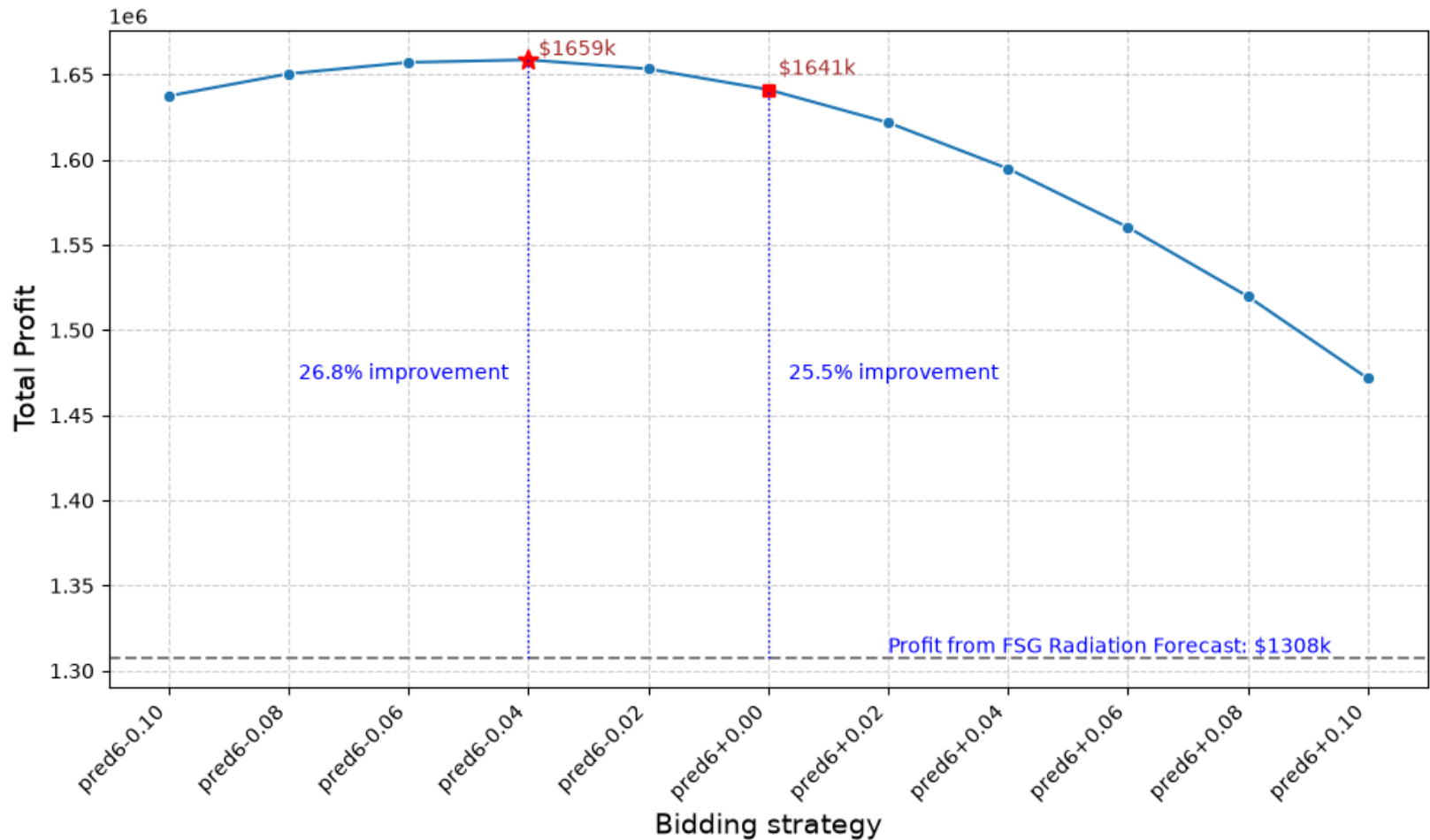
| | Forecasting Tool | | |
|-------------------------|------------------|-----------------|-----------------|
| Annual Values (\$1,000) | Current Practice | Model 1 | Model 6 |
| Revenues | \$ 4,200 | \$ 4,205 | \$ 4,356 |
| Production Costs | \$ 2,429 | \$ 2,429 | \$ 2,429 |
| Penalties | \$ 463 | \$ 454 | \$ 285 |
| Profit | \$ 1,308 | \$ 1,323 | \$ 1,641 |

- Using Model 6, profit increases 25% (from \$1,308,000 to \$1,641,000)

Can we do even better?

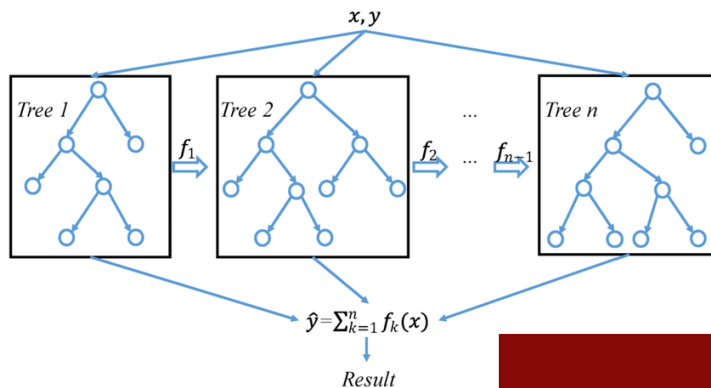
- $\text{OSR}^2 = 0.73$ is pretty good, but perhaps it can be further improved...
- Can we improve the forecasting?
- Can we improve the day-before contract *decision-making*?
 - Asymmetric cost structure of over/under-supply penalties.
 - Consider solar radiation as a “newsvendor-type model”

Beyond Predicting the Mean

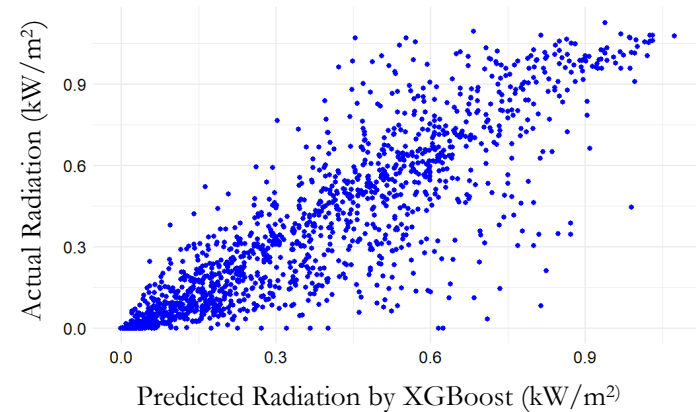


More advanced models you will learn

We can run **XGBoost**
(an ensemble model with boosting)



Scatterplot of Predicated Radiation versus Actual Radiation



| | R^2 | OSR^2 |
|---------|-------|---------|
| Model 6 | .73 | .73 |
| XGBoost | .86 | .79 |

- Over the next several weeks, we will see many different predication modeling tools, each with their own advantages and disadvantages